Research Title: Comparative Analysis of Single and Hybrid Machine Learning Models for Supply Chain Sales Forecasting

by

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ABSTRACT

Accurate sales demand prediction plays a vital role for efficient supply chain management, allowing for improved control of inventory, cost savings, and satisfied consumers. Conventional models typically fail to handle the complexities and diversity of real-world retail data. Machine learning models, especially hybrid ones, seem like a good option, but there isn't much research that compares them in new markets like Bangladesh. The research looks at five single ML models: Artificial Neural Network, Multi-Layer Perceptron, Gradient Boosting, Random Forest and Support Vector Regression (SVR). It also applies to ten hybrid combinations, such as RF+SVR and GB+RF, etc. We used a real-world retail dataset with 1,826 daily sales records and preprocessed the data by utilizing winsorization, seasonal decomposition, and feature scaling. To assess the performance of the models, four different statistical matrices were used: RMSE, MAE, MSE, and R^2. The top-performing single model was Random Forest (RMSE: 4.0998, MAE: 3.3014, MSE 16.8087, R^2: 0.5476), whereas ANN and MLP scored poorly (RMSE: 5.2962, R^2: 0.2450). Hybrid models have consistently performed better than single models. The best hybrid model, RF + SVR, had the best accuracy (RMSE: 4.0602, MAE: 3.2634, MSE 16.4851, R^2: 0.5563) and was more accurate in capturing complex sales patterns.

The results that hybrid ML models generate sales forecasts are far more accurate than single models. Our study emphasizes the importance of local data and hybrid models in developing reliable predictive tools for supply chains in emerging markets.

Keywords: Supply Chain, Demand forecasting; Machine learning; Hybrid models; Sales prediction; Random Forest; Support Vector Regression; Retail analytics; Artificial Intelligence

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LIST OF ABBREVIATIONS

AI Artificial Intelligence

ANN Artificial Neural Network

API Application Programming Interface

ARIMA AutoRegressive Integrated Moving Average

ARMA AutoRegressive Moving Average

CNN Convolutional Neural Network

CPU Central Processing Unit

CRM Customer Relationship Management

DBSCAN Density-Based Spatial Clustering of Applications with Noise

DD Demand Driven

DT Decision Tree

EDAS Evaluation based on Distance from Average Solution

ETS Error, Trend, Seasonality

FMCG Fast-Moving Consumer Goods

GB Gradient Boosting

GBT Gradient Boosted Trees

GH Grey Harmony

GLM Generalized Linear Model

GM Grey Model

GR Grey Relational

GRNN General Regression Neural Network

GUI Graphical User Interface

HA Historical Average

IQR Interquartile Range

KDE Kernel Density Estimation

KPI Key Performance Indicator

LR Linear Regression

LSTM Long Short-Term Memory

LU Logical Unit

MAD Mean Absolute Deviation

MAE Mean Absolute Error

MAPE Mean Absolute Percentage Error

MCDM Multi-Criteria Decision Making

ML Machine Learning

MLP Multi-Layer Perceptron

MLPR Multi-Layer Perceptron Regressor

MM Moving Mean

MQRNN Multi-Quantile Recurrent Neural Network

MSE Mean Squared Error

NN Neural Network

OSCM Operations and Supply Chain Management

RAM Random Access Memory

RBF Radial Basis Function

RF Random Forest

RMSE Root Mean Squared Error

RMSPE Root Mean Squared Percentage Error

RNN Recurrent Neural Network

SARIMA Seasonal AutoRegressive Integrated Moving Average

SBJ Special Business Jet

SC Supply Chain

SCM Supply Chain Management

SCP Supply Chain Planning

SCR Supply Chain Resilience

SCRE Supply Chain Risk Evaluation

SCRM Supply Chain Risk Management

SEM Structural Equation Modeling

SGD Stochastic Gradient Descent

SKU Stock Keeping Unit

SMAPE Symmetric Mean Absolute Percentage Error

SND Standard Normal Distribution

SQL Structured Query Language

STEM Science, Technology, Engineering, and Mathematics

STM Short-Term Memory

SVM Support Vector Machine

SVR Support Vector Regression

TI Technological Innovation

TSB Technical Service Bulletin

WAPE Weighted Absolute Percentage Error

WNN Wavelet Neural Network

XGB Extreme Gradient Boosting

CHAPTER 1

INTRODUCTION

1.1 Overview

This chapter provides a quick overview of the background of the supply chain demand forecasting, problem statement, and scope of our proposed research. After that, it discusses a few research questions and gaps in our study. It also outlines our proposed research objectives and contributions. The chapter ends with an outline for the thesis book.

1.2 Background Study on Supply Chain Demand Prediction

Supply chain demand prediction is one of the key issues in operations management, which ensures timely delivery and cut costs. In the modern business world, with its volatile patterns of demand, rapid technological shifts, and globalization of competition, the shortcomings of conventional forecasting models are becoming more and more obvious. (Vikram et al.2024)

Demand prediction is crucial for optimization supply chain and inventory in the retail sector. By precisely forecasting client demand, it assures the ideal amount of inventory to satisfy customer demands and prevent surplus inventory. This procedure is particularly essential when market conditions and customer needs are changing.(Praveen et al. 2024). Supply chain management relies heavily on demand prediction, which impacts customer retention, cost reduction, and operational effectiveness. Conventional approach, as like exponential smoothing and moving averages, find it difficult to take

into consideration the complexity of current markets. Artificial intelligence-powered models, like deep learning and time series analysis technique, are perfect for unstable markets because they can continuously update forecasts based on real-time data and incorporate greater variables. (Muthukalyani et al. 2023)

1.3 Problem Statement of the Study

After Most supply chain forecasts that integrate AI and machine learning face limitations in significantly improving demand planning. Traditional forecasting techniques are often overly reliant on historical data, making them less adaptable to real-time changes.(Kamal et al. 2023). This inefficiency becomes particularly problematic in industries with complex and dynamic supply chains, where accurate demand forecasting is critical to manage inventory optimization, cost reduction, and client expectations.

(Douaioui et al. 2024) AI and ML offer transformative potential by leveraging advanced algorithms able to process huge quantities of data, recognize complex patterns, and generating faster and more accurate predictions. Addressing these limitations is essential for enhancing operational effectiveness, flexibility in response to quickly shifting marketplace circumstances, and maintaining competitiveness in global markets. This challenge is especially pressing in the current era of digital transformation, where supply chains are increasingly disrupted by global crises and rapid shifts in consumer behavior. (Praveenadevi et al.2023). Failure to address these inefficiencies can lead to enhanced operational costs, resource wastage, diminished client retention By integrating AI-driven forecasting methods, organizations can bridge significant knowledge gaps in supply chain management, uncover actionable insights, and lay the foundation for further innovation in demand planning. (Tamanampudi et al.2024)

1.4 Scope of the Study

To ensure productivity, lower expenses, and retention of customers, supply chain management is an essential component of global businesses. It is essential to have reliable forecasts of demand to ensure the right quantity of inventory is on hand and there are as few issues as possible. Conventional prediction approaches rely highly on historical data and are unable to prepare for real-time changes in the markets. When worldwide crises are unpredictable, customer behavior is constantly changing, and the market is constantly changing, ineffectiveness is even worse in complicated supply chains. (Sahu et al.2021)

Combining AI and ML provides us new ways to address these problems. With the help of advanced methods, they can manage massive volumes of data, observe patterns, and make accurate forecasts in real time.(Gayam et al 2024).

This study examines how these technologies can be used globally to improve supply chains across industries and regions. The key objective of the research is to help close the knowledge gap around the world by providing useful information and a base for new ideas in forecasting the supply chain.

a) **Economic Impact:** Improved demand forecasting leads to a reduction in operational inefficiencies, cost cutting, and maintaining inventory at optimal levels for improved financial performance.

b) **Technological Advancement:** The integration of AI and machine learning will enable businesses to maintain global competitiveness by enabling data-driven decision-making and the adaptation of supply chain procedures in fluctuating market environments**.**

c) **Sustainability**: Precise demand forecasting reduces overproduction and waste, adding to the environmental sustainability of modern supply chains.

d) **Research Gap:** Whereas literature highlights the opportunity of AI in supply chain forecasting, the research gap concerns its application across diverse supply chain contexts, hence a need for further investigation.

1.5 Research Gap

(a) Lack of studies focused on comparative evaluation of both single and hybrid models.

(b) Limited research on retail data from Bangladesh.

(c) The lack of publicly available datasets from Bangladeshi retail companies.

1.6 Research Questions

After looking at a lot of current studies on supply chain demand forecasting, we came up with the following list of questions to ask when doing our thesis research.

(a) How can machine learning enhance the accuracy of demand forecasting in supply chain management?

(b) What are the most effective machine learning models to predict demand in supply chains?

1.7 Objectives of the Study

The following are the primary goals of the work to be done on the study:

a) To examine how machine learning may increase the accuracy of demand forecasting in SCM.

b) To assess the performance of various machine learning algorithms in demand prediction.

c) To identify the most effective ML approach for improving prediction accuracy in supply chains, both single model & hybrid mode

1.8 The Key Contributions of This Work

The following are the primary contributions to the research:

a) To fill in the gaps in demand forecasting research that lacked sufficient local datasets, we have been used real data from a Bangladeshi retail company.

b) We compared single and hybrid machine learning algorithms to identify the most efficient method for predicting sales.

c) To ensure that the comparison of model performance have fair and correct, standard evaluation measures (RMSE, MAE, MSE, R^2) has been used.

d) This study focused on the retail industry in Bangladesh, which has filled a gap in supply chain forecasting for emerging nations.

e) Encourage future study by using local data to forecast demand of sales with machine learning.

1.9 Outline of the Book

There are a total of six chapters to this thesis book. **Chapter 1**, which includes descriptions of the research's objectives, gaps, research questions, and other areas, provides a general overview of supply chain demand forecasting. **Chapter 2** gives a brief overview of previous studies on supply chain demand forecasting and the background research needed for the proposed work. In **Chapter 3,** we present a detailed description of the tools and techniques utilized in our research. **Chapter 4** gives a detailed discussion of developed architecture. **Chapter 5** completes the analysis of the results. In **Chapter 6,** research limits and future directions are discussed.

1.10 Summary of the Chapter

This chapter highlights the benefits of using various approaches and provides a brief description of supply chain demand prediction. The potential benefits of machine learning to improve demand prediction have also been discussed in this chapter.

CHAPTER 2

LITERATURE REVIEW

2.1 Overview

This chapter provides a comprehensive overview of the role of artificial intelligence (AI) in supply chain management. It examines the evolving landscape of AI applications in various supply chain functions, including demand forecasting, inventory optimization, manufacturing and retailing, logistics, and supplier management. Additionally, it discusses the relevant research work that supports us in executing the work in a more convenient way.

2.2 Existing Works

In conducting this research, it was discovered that there are a lot of publications from a variety of publishers, like Springer, Taylor & Francis, Elsevier, and others, which discuss how artificial intelligence will help supply chain management. For our literature review, we selected 43 research articles spanning the years 2016 to 2025. The number of papers corresponding to each year we used for our literature review is shown on **Fig.2.1.**

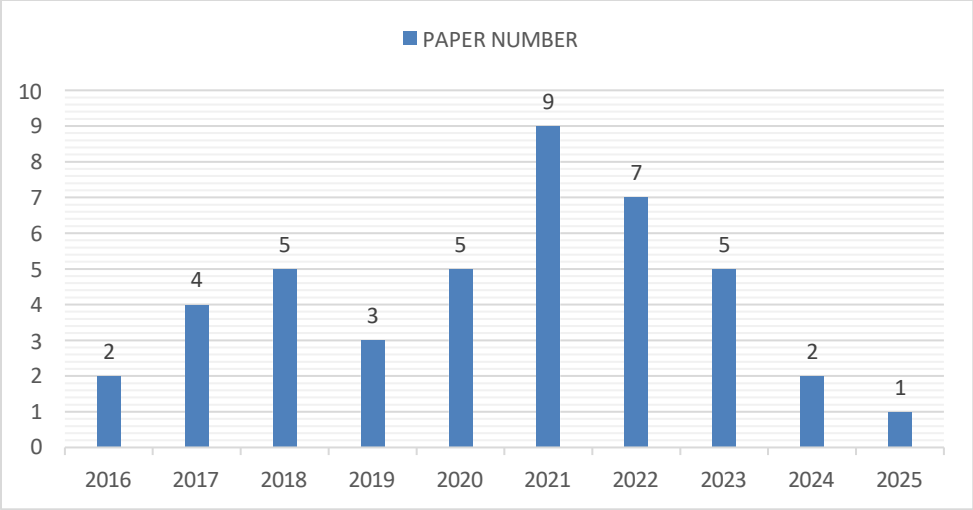


Fig. 2.1. The number of papers corresponding to each year we used for our literature review

2.3 Contextual Literature Review

Supply Chain Management is a complicated procedure that entails coordination with and managing numerous tasks, from sourcing raw supplies and shipping complete goods to customers. Integrating artificial intelligence (AI) technology into supply chain management can greatly increase efficiency, reduce expenses, improve decision-making, and adapt to changing demands on the market. By highlighting significant findings, approaches, and difficulties experienced by practitioners and scholars, this review of the literature intends to examine the present situation of AI study and implementation in SCM.

2.3.1 Definition of Supply Chain Management

The term "supply chain management" describes the organization and supervision of all production-related tasks. sourcing, procurement, logistics, and distribution of goods or services. The primary goal of SCM is to optimize the flow of materials, data, and financial resources throughout the whole system of supply chain, from raw material to end customers. Effective SCM ensures that Goods are

supplied to clients in a timely, and at the lowest cost while maintaining quality and customer satisfaction.

2.3.2 Definition of Artificial Intelligence

It refers to the discipline of developing intelligent machines, especially advanced computer algorithms. It refers to the equivalent objective of applying computers to learn about the intelligence of humans; however, AI has no obligation to constrain itself to physiologically observable approaches. (McCarthy et al. 2007)

2.3.3 AI Driven Demand Forecasting in SCM

Artificial intelligence-based demand forecasting systems have shown tremendous promise in enhancing supply chain performance. The qualities of the sector and the product's attributes are typically key factors in predicting demand for the items (Kolassa et al. 2016). It may be easier to predict the demand for functional goods near the customer as well as downstream of a supply chain than to attempt to determine the need for functional goods upstream of a chain. (Syntetos et al. 2016). As businesses grow into new markets or channels, machine learning-based prediction driven by predictive analytics has assisted them in improving customer interaction and producing more accurate demand forecasts than traditional demand forecasting techniques (Bohanec et al. 2017). The efficiency of these strategies for enhancing supply chain performance and compensating for the inconsistent success of ML-based prediction techniques, however, is not well supported by research (Makridakis et al. 2020).

2.3.4 AI in Decision Making in SCM

Artificial intelligence (AI) can be widely defined as the use of expertise systems, scheduling and organizing techniques, and simulations in supply chain decision-making. Agent-based computing technologies are among the most efficient techniques for modeling the relationship of system components and evaluating effectiveness in practical life.(Abar et al.2017) Expert systems, also known as systems built on knowledge, are another form of artificial intelligence that encompass various fields and techniques of problem-solving with the aim of enabling systems to perform operations that are normally performed by human experts. (Zarbakhshnia et al., 2018) .According to (Abedinnia et al. 2017) 's study, recent AI advancements have not only made it easier to recognize and foresee interruptions that could affect regular system functions including fraud identification, preventative maintenance, and system breakdown etc. it also assisted in faster and more data-based recovery of the system.

2.3.5 AI in Smart Manufacturing and Retailing

AI helps manufacturers and retailers improve the management of supplies, seasonal offers, selection, and pricing. Additionally, it improves real-time forecasting accuracy. Semantic division and recognition of objects' efficiency, or the ability to categorize product types, such as differentiating an instrument from a component, have significantly increased recently. (Wen et al. 2018). AI is used by manufacturers to improve KPI and examine them in real time. Developing a model with virtualized programs, for example, helps with consumption of energy management and enhanced forecast, evaluation, and mitigation of material and personnel limitations. More importantly, it recommends solutions and alerts the professionals to any problems.

According to (O'Reilly et al.2019), AI-assisted production not only enhance assembly line process but also deduct cost, limit waste, and sped up time to marketplace. By delivering personalized and convenient shopping experiences, AI has allowed merchants to boost both the quantity of consumers, and the average dollar amount they purchase. Marketers now have more knowledge of what consumers desire, prior to their clients apparently knowing it. (Deb et al. 2018). To produce customer-oriented products, intelligent AI and human beings have been working together to mass-create goods. Manufacturing facilities all over the world have benefited from AI in supply value chains that are more connected & engaged (Klumpp et al. 2018).

2.4 Theoretical Literature Review

(Hasija & Esper, 2022) explore how organizational issues balance real implementation and utilization of AI with any prospective advantages for SCM in their study. They propose strategies to highlight AI reliability via thematic analysis and leader surveys. The findings explain managerial implications for AI reliability in SCM and contribute to academic discourse on technological adoption and digitalization. (Farheen et al. 2021) explore the role of artificial intelligence (AI) in a sustainable and resilient supply chain during the COVID-19 in their study. The review employed systemic subject modeling and bibliometric evaluation to find research patterns in 162 articles from the SCOPUS repository. The outcomes indicate a way for AI research in supply chain resilience to assist technology in advancing and lowering risks. This study presents supply chain managers advice on how to make their operations stronger and suggestions for further study. By performing a concise review of existing literature, , (Hieu et al. 2020) identify ways in which AI supports SCM. The study further discusses the main areas of SCM

where AI could be implemented and looks at the past uses of AI in SCM that have been advantageous. The main conclusions of this study suggest that a system built on agents is among the most often used AI approaches for solving various SC issue aspects.(Erfan et al. 2021) explore the applications of ML in SCM using a conceptual framework in their research paper. ML techniques contribute to supplier selection, manufacturing, inventory control, shipping, delivery, evaluation of risks, demand estimate, sustainable growth, and the circular economy. Given that most AI techniques in SCM are only useful for well-structured operational and tactical challenges, this article recommends looking into real-time pricing applications. (Milan et al. 2021) In their research paper, it has been shown that AI applications in Enterprise Information Systems, focusing on CRM, SCM, the management of the product lifecycle, accounting, finance, manufacturing planning, warehousing and shipping, and employee relations. Decision-making and robotics have been carried out using machine learning methods in the context of the expected AI services to enhance capabilities.

(Vahid et al. 2020) introduce a framework for genetics and an AI method based on biological evolution to predict and estimate export sales. It performs a six- week sales prediction, examines an empirical instance of an export business, and gives a variable sensitivity evaluation for the causal forecasting model. AI-based technologies and conventional sales and demand forecasting strategies are contrasted to revealing the advantages and disadvantages of each. (Bernd Hellingrath et al. 2019) emphasize recognition approaches for supply chain execution activities and give an overview of AI methods and SCM tasks. This study draws attention to the wide variety of AI approaches and difficulties in SCM and

logistics. Additional applications could be explored in further research, along with the applicability of AI to issues with supply chain execution.

The objectives of (Oana et al. 2020) research is to develop a performance evaluation approach that connects particular issues to appropriate supply chain management KPIs. Data mining establishes the connection between issues and KPIs, and a GUI built around multilayer perceptron AI and neural networks forecasts the most reliable KPIs for every issue. (Samuel et al. 2021) highlight business applications and useful findings that demonstrate the connection between operations and supply chain management (OSCM) and AI in their study. It provides research recommendations for academics and practitioners while introducing a few chosen articles. The objective of this paper is to increase understanding of OSCM-related AI applications and findings.

(Mehrdokht et al. 2021) In this research, they evaluated SCM approaches to decision-making as well as recognized research areas. They also discuss emerging disciplines, including sensing/interacting and instruction. However, it fails to provide a comprehensive analysis of AI methods and how they correlate to SCM. Methods of learning have become growing more popular, but ways of sensing and interacting are still relatively new.

(Umamaheswaran et al. 2019) The team developed a demand prediction system that utilizes multilayer feedforward neural networks (FNN), backpropagation, and artificial neural networks (ANN). By minimizing supply-demand inconsistencies and controlling fluctuations in customer demand, this approach increases the revenue of the business. With an accuracy of 75% to 80%, 2 or 3% points more than the benchmark standard, the ANN algorithm increased SCM performance overall. (K. Nemati et al. 2017) provide an assessment of AI

methods for forecasting applied in service and non-service supply chains, focusing on the best practices. In airplane component supply chains, people have employed neural networks and MSE instead of more conventional methods like Croston, SBJ, and TSB. Their research indicates that NN with additional features lowers the cost effects of prediction of demand efficiency for irregular needs.

The research work by (George et al. 2018) reviews the literature on supply chain risk management using AI approaches. The terms, classifications, and uncertainty of supply chain risk have been studied in this paper. This research highlights limitations and fresh ground by organizing a collection of literature applying an AI approach. The study provides suggestions for future SCRM and AI study areas. (Amine et al. 2020) explore the effects of AI, supply chain adaptability, and supply chain capacity (SCP) in an evolving and uncertain supply chain system. They examine a method using SEM and suggest that while AI has a short-term impact on SCP, its capacity for assessing data should be leveraged to develop SCRes for long-term SCP. This study discusses the importance of AI capability in generating long-term SCP.

In this research (Petri et al. 2021) explore SCM and AI concepts and identify supply chain studies and applications based on AI. They assess the methodologies and principles across multiple new AI-based business strategies from diverse companies. This paper identifies prospects for value creation and proposes a way for developing enterprise strategies for AI supply chain deployments.

The purpose of the research conducted by (Amine et al. 2021) is to identify patterns in AI techniques for developing supply chain resilience (SCRes) methods using a hybrid multi-criteria decision-making (MCDM) technique that involves AI-based algorithms such as fuzzy networks, wavelet neural networks (WNN), and the evaluation based on distance from average solution (EDAS). Information from 479

industrial companies identifies fuzzy logic programming, machine learning, big data, and agent-based applications as potential SCRE models. They present an extensive process for creating SCREs for specialists in AI implementation.

In their paper, (Youssra et al. 2021) propose a structure for making choices regarding the best way to implement AI in the supply chain. The main contribution of the study includes their classification, which facilitates the development of a structure for AI-driven SCs and clarifies the role of AI in supply chains. Their research highlights for both academics and professionals the factors to consider while applying AI, especially conceptualizing and developing intended objectives.

(Pawankumar et al. 2023) investigate the use of AI in smart supply chain organization to close a research gap in smart source chain management. They discuss Microsoft 365 and artificial intelligence ideas for improving SCM performance. This research looks at Facebook's deployment, production, and marketing applications of AI. The research examines the benefits, characteristics, and potential of Microsoft 365 for effective SCM.

2.5 Related Works

In this research study (Yasaman et al.,2022), they used a retail store dataset and employed both traditional models, such as ARMA, ARIMA, and SARIMA, as well as various advanced deep learning models, including Prophet, LSTM, and CNN. They evaluated performance using RMSE and MAPE to evaluate the outcome, and they selected the supeior model for forecasting. While the outcomes for Prophet and CNN also show satisfactory performance, their results show that the stacked LSTM model beats the other approaches.

They (Imece et al., 2022) ensemble four types of models—Holt-Winters, Ridge Regression, Random Forest, and XGBoost—to forecast better outcome and lower error rates. They created binary, triple, and quadruple combinations of those models: the outcome shows that the individual model's lowest error rate (RMSPE) was Random Forest with 15.7% RMSPE; in the ensemble, the Holt-Winters & XGBoost models combination achieved the lowest 10.7% RMSPE. This study (Rathipriya et al,. 2023) shows the performance in a weekly sales dataset of a pharmaceutical company using shallow and deep learning models. They are defined by deep neural networks like STM and Stacked LSTM, as well as shallow neural network models like GRNN, RBF_NN, and P_NN. They use RMSE to evaluate the demand forecasting model, and the results show that the shallow neural network model had a lower error rate of 6.27 compared to the deep neural network.

Their study (Zhang et al., 2024) introduces an innovative Hybrid Attention-based Long Short-Term Memory (HA-LSTM) network for time series forecasting, especially for predicting customer demand in retail. The model does an effective job of capturing long-term dependencies and regional trends by using LSTM layers with multi-head self-attention. On a massive dataset, HA-LSTM improves MAPE by 15% and reduces RMSE by 12% when compared with traditional methods. By means of attention weights, this study shows the model's interpretability, so improving knowledge of time relevance in forecasts and so supporting developments in deep learning for time series analysis.

This paper (Islam et al.,2024) compares various machine learning methods, including GB, AdaBoost, RF, XGBoost, ANN, and a new hybrid architecture (RF-XGBoost-LR), for real-time sales data analysis. The hybrid model, which combines

bagging and boosting techniques, shows greater accuracy and dependability in forecasting supply chain requirements and demand. The outcome suggests it involves industry-specific modifications to enhance client satisfaction and decision-making by means of exact demand prediction.

In this study (Amrutkar et al., 2022), they look at how smart decision analytical systems combine decision analysis with predictive modelling to enhance the accuracy of business need forecasts. It shows how data mining can provide insights from large datasets, outside predicting limitations. Compared to Decision Tree and Generalized Linear Models they give 71% and 64% accuracy; GB model is the best performing forecasting model with 98% accuracy. Findings show that machine learning can manage vast amounts of data and help people make better decisions, even though it can be hard to analyze. They (Taparia et al., 2023) have developed a hybrid model for enhancing forecasting accuracy using historical demand data and SKU pricing. The Random Forest regressor had the great outcome, with an MAPE of 8% when evaluated at 1000 SKUs. They apply a hybrid model and find that this number dropped to 7.74%. They find that hybrid models are more efficient than individual regression algorithms. Their study also recommends including more elements to manage the constraints resulting from a small set of parameters for future work. This paper (Abhinaya et al., 2023) shows better accuracy than conventional techniques by means of a prediction model based on the XGBoost algorithm. It underlines operational efficiency and retail profitability's need for accurate sales forecasting. Using data analytics to improve inventory control and sales forecasts, large supermarkets can spot trends and outliers to project future sales volume. This study also investigates machine learning principles and variables that impact sales forecasts.

In their study (Taşdemir et al., 2022) they examine that intelligent machine

learning techniques Grey Model (GM), ANN, and SVM to forecast the short-term demand for a food supplement, that rapidly grew during the pandemic. This study found that after optimizing parameters and analyzing historical data, all the models performed effectively. GM (1,1) performed somewhat better than the others. The results suggest that businesses may make their forecasting strategies more effective by applying these smart methods to a wider range of items, which would lead to higher demand. This paper (Filali et al.,2022) shows a deep-learning technique for predicting demand using LSTM, which proves that it performs more effectively than traditional methods like ARIMA and exponential smoothing. The improved model had a RMSE of 4487.32 and a symmetric mean absolute percentage error (SMAPE) of 0.026. Researchers noticed that LSTM may assist SCM be more accurate and effective at making predictions. The finding has significance for future applications and the development of hybrid models.

This research (Firat et al., 2021) examines changes in air travel as well as how big data analytics can be used to boost demand for airline trips. Including ANN, LR, GB, and RF, it provides a novel method for measuring ideal seat demand by applying ML models. This study used a detailed World Bank dataset from before COVID-19 (2014–2019) and found that the GB method had the highest prediction accuracy, with a MAPE of 6.741%. Further research will focus on how seasonality and ticket costs influence passenger demand.The researchers (Firat et al., 2025) applied sales data to assess multiple machine learning approaches, such as MLP, Multi-Horizon Quantile Recurrent Neural Network (MQRNN), and RF, to see which model could best predict demand for FMCG. After analyzing all the models, they determined that MQRNN did the greatest for item B in August, while RF did very well in July. The research demonstrates that the effectiveness of the model changes based on the type of product and the time predicted. The study illustrates

how important it is to consider these factors when making predictions.

The study by (Panarese et al.,2022) demonstrates that XGBoost effectively predicts customer demand, as proven by multiple error metrics including MSE, MAE, MAPE, and WAPE, resulting in enhanced accuracy in predicting sales. The XGBoost regressor has been shown to outperform traditional GB regressor algorithms for both existing and new products. A prototype platform has been developed to validate the complex logistics management of trading businesses.

This study (Nassibi et al.,2023) examines how sales in the food market change when the economy is unstable around the world. They also stress how important it is to use techniques like Support Vector Machine (SVM) and Long Short-Term Memory (LSTM) models to make better guesses about how much candy would sell in Saudi Arabia. The results indicated that LSTM achieved better than SVM, minimizing errors by as much as 77%. This finding shows that more research will be necessary to learn more about promotional sales and see how the COVID-19 scenario has affected sales patterns.

Table 2.1: Summary of Existing Works Related to Supply Chain Demand Forecasting

Model	Dataset	Performance	Statistical Tools	Source
SARIMA, ARIMA, LSTM, CNN, Prophet	Pecan Street Dataset	LSTM	RMSE MAPE	(Yasaman et al.,2022)
Holt Winters, Ridge Regression, RF, Xgboost.	Global pharmaceutical company in Türkiye	Individually Random Forest with 15.7% RMSPE value, and 10.7% RMSPE value in Holt Winters & Xgboost models combination	RMSPE	(Imece et al., 2022)
ARIMA, RBF_NN, GR_NN, P_NN, LSTM, Stacked LSTM	Pharma sales data	GRNN, RBF_NN, P_NN	RMSE	(Rathipriya et al,. 2023)
HA-LSTM, ARIMA, Prophet, and Vanilla LSTM	Predict Future Sales	HA-LSTM achieved a 15% improvement in (MAPE) and a 12% reduction in (RMSE).	RMSE, MAE, MAPE	(Zhang et al., 2024)
XGBoost, ANN, GB, AdaBoost, hybrid framework (RF-XGBoost-LR)	US retailer dataset	(RF-XGBoost- LR),	MAE, MSE, R2	(Islam et al.,2024)

Table 2.1 continued from previous page.

Generalized Linear Model (GLM), DT, Gradient Boost Tree	E-fashion store Dataset	Gradient Boost Algorithm	RMSE MSE MAE	(Amrutkar et al., 2022)
SVR, Polynomial Regression, Decision Tree Regression, Random Forest Regression	Retail Store Dataset	Random Forest Regressor	MAPE MAD MSE	(Taparia et al., 2023)
XGBoost, Lasso Regression, Ridge Regression	Supermarket Sales Dataset	XGBoost	MAE RMSE R2	(Abhinaya et al., 2023)
GM (1,1), ANN, SVM	Türkiye Food Supplement Sales Dataset	GM (1,1)	MAPE	(Taşdemir et al., 2022)
ARIMA, ETS, RNN monolayer, RNN multilayer, multilayer LSTM	Moroccan Pharmaceutical Product Sales Dataset	Multilayer LSTM	SMAPE, RMSE	(Filali et al.,2022)
ANN, LR, GB, RF	Turkish Airlines Dataset	GB	MAPE	(Firat et al., 2021)
MLP, Multi-Horizon Quantile Recurrent Neural Network (MQRNN), RF	Daily Sales Data for FMCG	MQRNN	MAPE	(Firat et al., 2025)

Table 2.1 continued from previous page.

GB, XGBoost	Different Store Sales Data	XGBoost	MAE MSE MAPE WAPE	(Panarese et al.,2022)
LSTM and SVM	Confectionery company in the Kingdom of Saudi Arabia	LSTM	MAPE, RMSE	(Nassibi et al.,2023)

According to the reviewed literature, machine learning and hybrid models have become increasingly important in improving the accuracy of sales forecasts across various sectors, including retail, food, fashion, pharmaceuticals, and transportation. Researchers have studied various models, including ARIMA and Holt Winters, deep learning models like LSTM and GRNN, and hybrid models like RF-XGBoost-LR. Those models perform differently across datasets in terms of RMSE, MAE, MAPE, and R^2. The literature review shows that ML models such as Multilayer Perceptron (MLP), Artificial Neural Networks (ANN), Gradient Boosting (GB), Random Forests (RF), and Support Vector Regression (SVR) are widely used for predicting sales demand with satisfactory outcomes.

The literature also shows that hybrid models are more accurate than single models because they combine the strengths of numerous models to better address complicated patterns in sales data. The primary objective of this study is to compare and measure the effectiveness of every single model (MLP, ANN, GB, RF, and SVR) and then create hybrid models according to pair-wise comparisons like MLP+ANN, MLP+GB, MLP+RF, MLP+SVR, ANN+GB, ANN+SVR, ANN-RF, GB+RF, GB+SVR, and RF+SVR. We will use commonly used metrics such as Mean Squared Error (MSE),

Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Coefficient of Determination (R^2) to analyze how well each of the models performs and then pick the most effective one for sales forecasting.

2.6 Summary of the Chapter

This chapter gives a full review of the different tools and methods that can be utilized for predicting demand in the supply chain. It also shows how artificial intelligence has improved in various aspects of the supply chain.

CHAPTER 3

BACKGROUND TOOLS AND TECHNOLOGIES

3.1 Overview

This chapter goes into more detail on the strategies and approaches that have been utilized in the research process for developing the suggested model.

3.2 Tools

This section presents a brief description of every tool that will be required to execute the researched procedure.

3.2.1 Machine Learning Libraries

Machine learning projects generally applied Python libraries such as Pandas, NumPy, Matplotlib, Sklearn, Seaborn. The proposed approach uses these types of libraries.

3.2.1.1 Pandas

A Python programming language package called Pandas helps in data research and analysis, enabling programmers to construct high-level data structures that are understandable and accessible. Pandas collaborate to benefit various areas, including science, statistics, finance, and engineering. The Pandas library is helpful

for this purpose, as it provides various data structures, functions, and components that support data extraction and processing tasks. Pandas employ data frames to manage machine learning tasks. Spreadsheets and SQL tables utilize 2D data frames. It improves system efficiency by helping developers analyze data. With the support of the Pandas toolkit, experts can organize, categorize, restructure, and oversee large datasets. (TutorialsPoint, 2025)

3.2.1.2 NumPy

Numerical Python (NumPy) is a free and open-source Python library designed for scientific and numerical calculations. This library provides numerous mathematical operations, enabling calculations on arrays with multiple dimensions and matrices. Researchers widely use this approach to analyze data. NumPy makes it possible for developers to build intelligent systems that react in real time to real-life scenarios. This is possible because of the library's complex Fourier transform, linear algebra, and random number generation methods. The NumPy library performs mathematical operations on multi-dimensional arrays. Python algorithms exhibit improved execution speed, which improves overall system productivity. Developers may engage with the NumPy community to address technical challenges and subsequently resolve these issues. This feature allows programmers to quickly solve problems. ((*NumPY Documentation*, n.d.)

3.2.1.3 Matplotlib

Matplotlib is a well-known Python graphical toolkit that works well with NumPy and provides an object-oriented API for displaying plots in a variety of GUI programs, including TkInter, wxPython, Qt, and GTK. John D. Hunter created Matplotlib, which is used in physics, astronomy, engineering, and biological research for data visualization. A large community of users has contributed to it. It has helped researchers construct publication-quality graphs for major scientific

projects, including the Event Horizon Telescope's first black hole picture visualization. STEM curriculum and online learning platforms use Matplotlib to teach programming and data visualization, particularly in Jupyter Notebook environments. It is a vital part of scientific and educational visualization processes, as demonstrated by its broad use in both academia and industry. (Wikipedia contributors, 2025)

3.2.1.4 Sklearn

Scikit-learn, also referred to as sklearn, is a well-known open-source Python package for data modelling and machine learning that originally became available in 2010. It includes a broad range of classification, regression, and clustering model, like DBSCAN, k-means, gradient boosting, RF, and SVM.A simple fit/predict process is made possible by sklearn, which provides a standardized and condensed interface for model forecasting and training and allows it to integrate seamlessly with other Python libraries like NumPy & SciPy. There are features for classification, regression, grouping, reducing dimensionality, selecting models, and data preprocessing in the library that support both supervised and unsupervised learning. It is an essential tool in the Python machine learning ecosystem for researchers and users due to its interoperability with data structures like Pandas dataframes and NumPy arrays, which improves its versatility. (*What Is Sklearn? | Domino Data Lab*, n.d.-b)

3.2.1.5 Seaborn

Seaborn is a widely used library for data visualization in Python that operates on top of Matplotlib. The library makes it easy to produce statistical graphs that are both appealing and useful. Its framework makes it easier to interact with Pandas Dataframes, which helps users visualize and understand data better.

Seaborn has a multitude of visualization capabilities, such as scatter plots, line plots, bar graphs, and heat maps. It also supports advanced statistical analyses like regression and distribution charts. One of its primary benefits is its ability to generate attractive plots with minimal coding, which can be supported by versatile color palettes and styles. Seaborn is also excellent at generating complex multi-plot graphs that are relatively simple to compare across different data sets or terms. Its adaptability makes it suitable for data exploration, evaluation, and display. Seaborn is a sophisticated and flexible Python data visualization tool that combines easy access with strong analytical tools. (*Datacamp*, 2023)

3.3 Techniques

This section describes the many techniques required to extract features, handle outliers, and scaling dataset features to execute the proposed model.

3.3.1 Winsorization for Outlier Handling

Winsorization is the process of substituting fewer extreme values for extreme values in a dataset. In contrast to trimming, which eliminates outliers, winsorization limits the extreme values at certain levels, thereby preserving all data points while reducing their influence on statistical parameters. This approach minimizes the influence of extreme values on statistical analysis and model training while preserving the data points (Da Silva, 2025). The parameters "lower_limit = 0.05" and "upper_limit = 0.05" provide the boundaries for winsorization. The `limits` parameter in the `winsorize` function accepts an array or list consisting of (lower, upper) values. These values show the percentage of data from the distribution's lower and upper tails that will be winsorized.

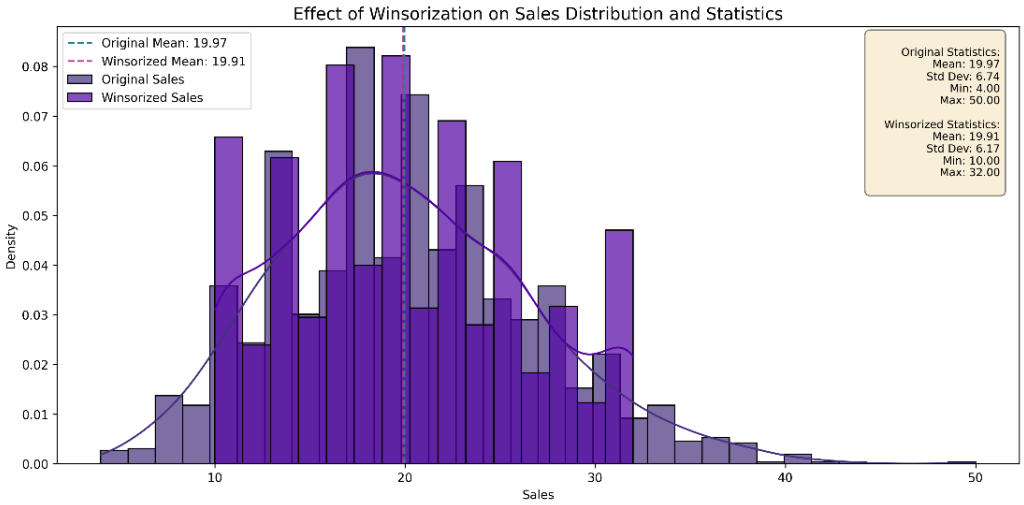


Fig. 3.1. Effect of Winsorization on Sales Distribution and Statistics

Table 3.1: Statistical Difference Before and After Applying Winsorized Sales

	Count	Mean	Std	25%	50%	75%	Min	Max
Original Sales	1826	19.97	6.74	15.0	19.0	24.0	4	50
Winsorized Sales	1826	19.91	6.17	15.0	19.0	24.0	10	32

Here, [0.05, 0.05] suggests the value at the 5th percentile will substitute for the values at the bottom 5%, and the value at the 95th percentile will substitute the values at the top 5%.Winsorization helps to minimize the impact of outliers while retaining them, thereby saving a greater portion of the data, which can be useful in all-time series models. Because "sales" in Sales are a time-series variable, and eliminating outliers may create gaps or change patterns over time. Winsorization offers a method that minimizes the influence of outliers when retaining the data points and keeping the temporal sequence. It is used for modifying numbers under the 5th percentile to the 5th percentile value and to the 95th percentile value for values beyond the 95th percentile. There is no change in the numbers between the 5th and 95th percentiles.

3.3.2 Data Scaling with StandardScaler

StandardScaler is a feature scaling technique used to normalize the values of numerical features which are adhered to the Standard Normal Distribution (SND). It changes the data in such a way that the standard deviation equals one and the mean equals zero. It is perfect for algorithms that consider normally distributed data, such as logistic regression, SVM, and NN. The procedure consists of two primary steps: fitting and transforming. Following feature engineering and data cleaning, the scaler determines the mean and SD (standard deviation) for every feature of the dataset throughout the fitting process. The transformation step removes every feature value from the mean. StandardScaler is normally used prior to training models, after splitting the data into testing and training sets, even if the offered code snippet does not explicitly implement it. According to the needs of the model, scaling of the target variable for example, the 'sales' column in regression tasks, may also be considered. (GeeksforGeeks, 2025)

$$StandardScaler\ z = \frac{x-\mu}{\sigma} \tag{3.1}$$

Here, $\mu = Mean$

$\sigma = Standard\ Daviation$

$x = Data$

3.4 Summary of the Chapter

This chapter provides a brief description of every tool and technology which applied in completing the proposed work.

CHAPTER 4

SYSTEM ARCHITECTURE AND METHODOLOGY

4.1 Overview

This chapter includes a summary of the technique used in developing the proposed framework. The experimental setup, exploration of the dataset using different techniques, data preprocessing process, overview of the usage of various algorithms in this study, and details of the work process discussed using model architecture, and model training and validation are among the topics discussed.

4.2 Dataset Description

For this research, we used a dataset derived from the study conducted by Jahin & Shahriar (2024). They gathered data from a well-known retail company in Bangladesh; however, Because of the unique nature of the dataset and confidentiality contracts with the retailer, the preliminary sales data cannot be published publicly., and the company's identity has not been disclosed. Consequently, they anonymized and aggregated this processed data to protect the retailer's confidentiality.

4.3 Exploration of Data Set

The dataset used in this study included 1,826 daily sales records for a certain product that have been carefully gathered to ensure that it was accurate and reliable.

Each entry has two main columns: a timestamp in the standard date format (YYYY-MM-DD) to make it simpler to analyze time series data and the number of units sold, which are shown as whole numbers that show how much of the product was sold each day. In this case, one sales unit is a group of 1,000 units. This dataset lacks other outside factors like the weather or macroeconomic factors, but the holiday data makes it easier for the model to accommodate fluctuations in demand that aren't consistent. The most important things we discovered through the dataset are that product demand implies a seasonal pattern, with sales being lower at the beginning and end of the year and much higher in the middle portion of the year.

4.3.1 Sales Data Seasonal Decomposition Analysis

This graph looks at sales data from 2013 to 2017 and breaks it down into three main parts: trend, seasonality, and residuals.

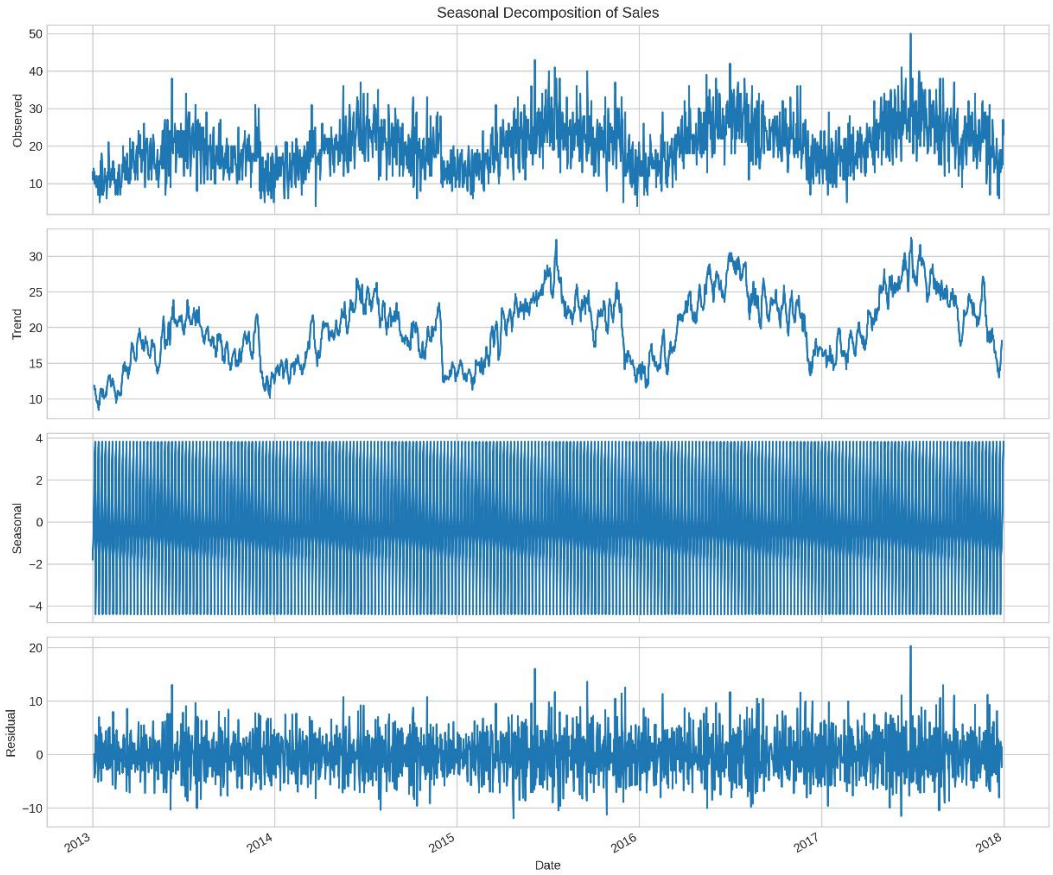


Fig. 4.1. Seasonal Decomposition Analysis of Sales Data

It is important to understand these parts in order to accurately analyze and predict data. The observed part shows actual sales numbers fluctuate a lot, with different highs and lows that imply seasonal changes. It showed a general upward trend until the middle of 2016, but then there was a tiny state of decline. This may indicate that the market has fluctuated or that customers have responded differently. The trend part indicates the long-term pattern of sales, which slowly increased from 2013 to the middle of 2016. But this trend levels out and then declines a little, which could mean that demand has begun to level off or that other economic reasons have slowed the growth of sales. The seasonal part reveals that sales have followed regular, cyclical patterns, which proves that the seasonal structure is strong. These patterns illustrate that prices go up and down at regular times, which is when people usually buy products during sales. The residual pattern varies near zero, showing that trend and seasonality capture most data signals. However, rapid sales spikes and falls may suggest short-term offers, unanticipated demand, or supply chain interruptions. These inconsistencies show that while the data set is consistent, external factors or occasional events still affect sales, requiring strong models.

The seasonal decomposition analysis concludes that long-term trends, predictable seasonal patterns, and random changes comprise the sales time series. This information is especially useful for making predictions and planning strategies.

4.3.2 Heatmap Time Series Plot

The heatmap analysis of monthly average sales from 2013 to 2017 shows seasonal and year-over-year sales trends. Darker colors indicate more sales, and each heatmap cell reflects the average sales for a certain month and year. The data

shows that sales peak in the summer, especially between June and August, with July being the best month of all five years.

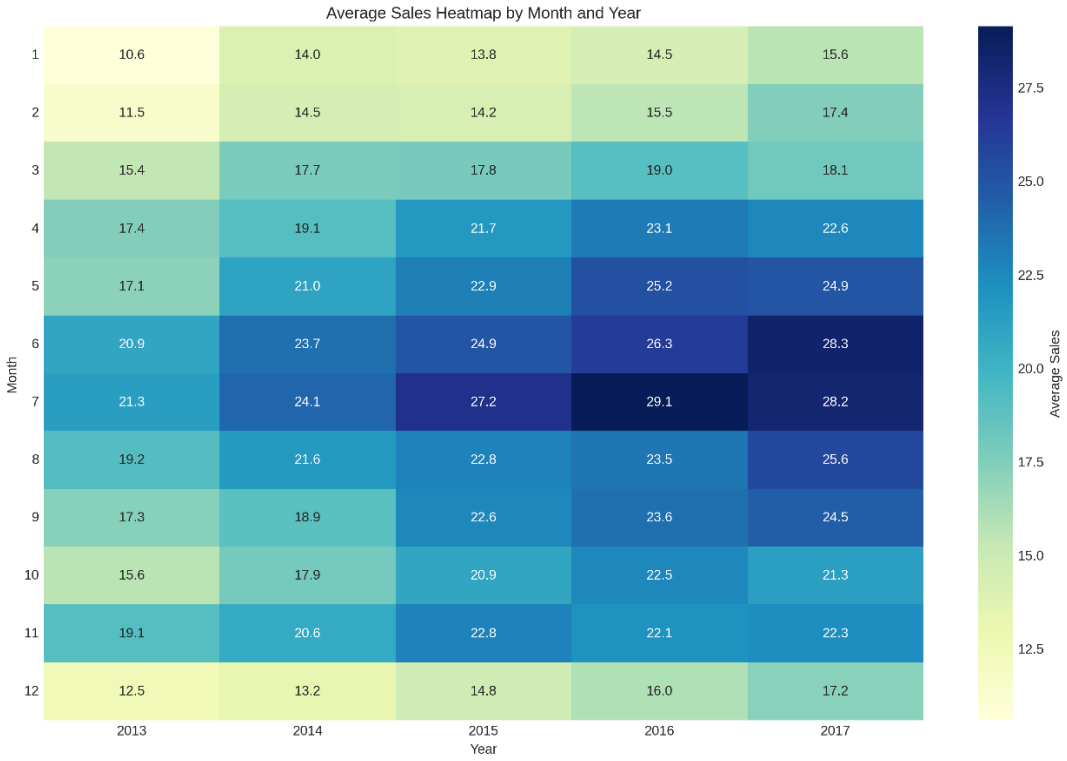


Fig. 4.2. Heatmap Analysis of Monthly Average Sales From 2013 to 2017

This regular July high shows strong seasonal demand, affected by mid-year promotions, holidays, or weather. In July 2016, sales averaged 29.1, the highest in the dataset, followed by July 2015 at 27.2 and July 2017 at 28.2. The data shows that average sales are lowest in January and December. January 2013 had the lowest average sales of 10.6, and December sales fell to 13.2 in 2014 and 12.5 in 2013. This tendency may indicate a seasonal slowdown or a decline in holiday sales. From 2013 to 2016, average monthly sales increased year-over-year, rising from 17.1 in May 2013 to 25.2 in May 2016. Sales growth tended to steady or drop in several months in 2017, suggesting market maturity or external difficulties. The accompanying heatmap shows strong seasonal sales trends and an overall growth trend, which will help develop forecasting models that account for seasonality and guide inventory and marketing strategies.

4.3.3 Rolling Mean and Approx. Confidence Interval

The graph "Sales with Rolling Mean and Approx. Confidence Interval" shows daily sales data from 2013 to 2017 and employs a 95% confidence interval and a 30-day rolling mean to point out significant trends.

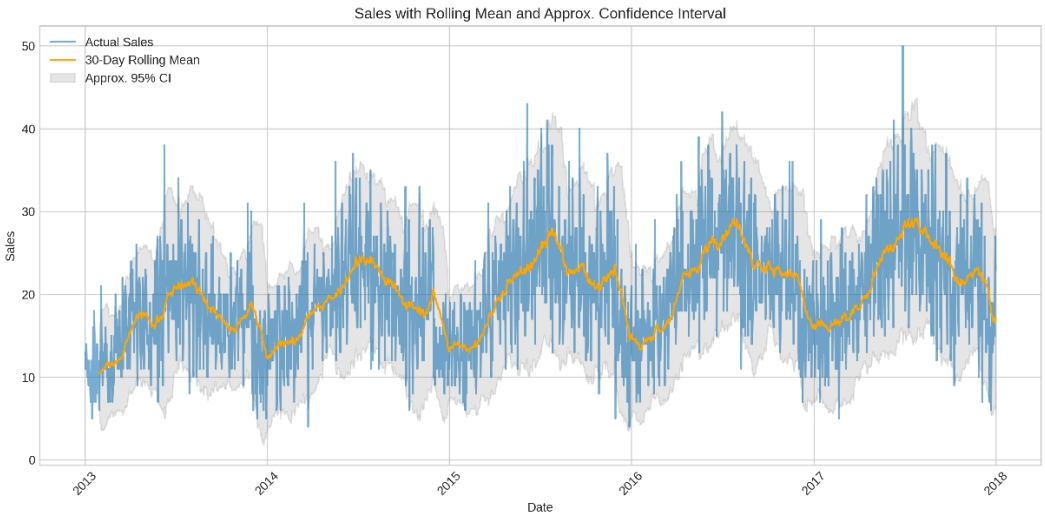


Fig. 4.3. Sales with Rolling Mean and Approx. Confidence Interval

A certain increasing trend and significant seasonal patterns with frequent annual highs and lows are revealed when the rolling mean (orange line) smoothes out the excessive volatility of the raw sales (blue line). The confidence interval, represented by the grey shaded area, shows periods when sales variability is larger and lower. It usually expands during high-activity seasons like the holidays. The above graph is very important for predicting, making plans, and managing inventories since it shows how trends, seasonality, and uncertainty affect sales behavior.

4.4 Data Pre-Processing

Data preparation is an important step in the process of machine learning or data analysis. It involves converting raw data into a format that can be used to train and test models. The major goals have been to make the data more effective, lower variance, and achieve the most feasible outcome from the next phases. The preprocessing step has been applied in this research to classify as follows:

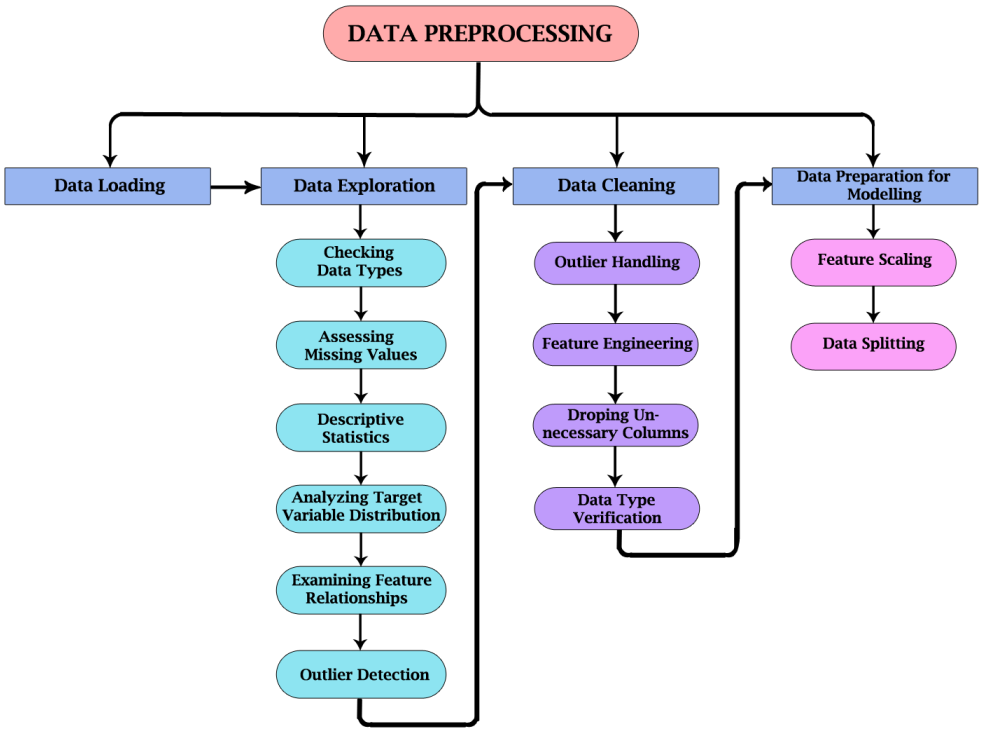


Fig. 4.4. Data Pre-Processing Steps

4.4.1 Data Loading

The first step has been to import the data from an Excel file named "dataset.xlsx" into a panda DataFrame. This file contains a structured representation of the data, making it simpler to modify and analyze.

4.4.2 Data Exploration

We assessed the dataset carefully prior to making any modifications. This consisted of the following:

4.4.2.1 Checking Data Types

It is important to know which type of data is in each column to make sure this fulfills research requirements, like transforming the "date" column into datetime objects.

4.4.2.2 Assessing Missing Values

We assess each column and figure out if any data is missing. Although there have been no missing data in the columns used for analysis, this step is essential for identifying which removal processes are needed in the datasets.

4.4.2.3 Descriptive Statistics

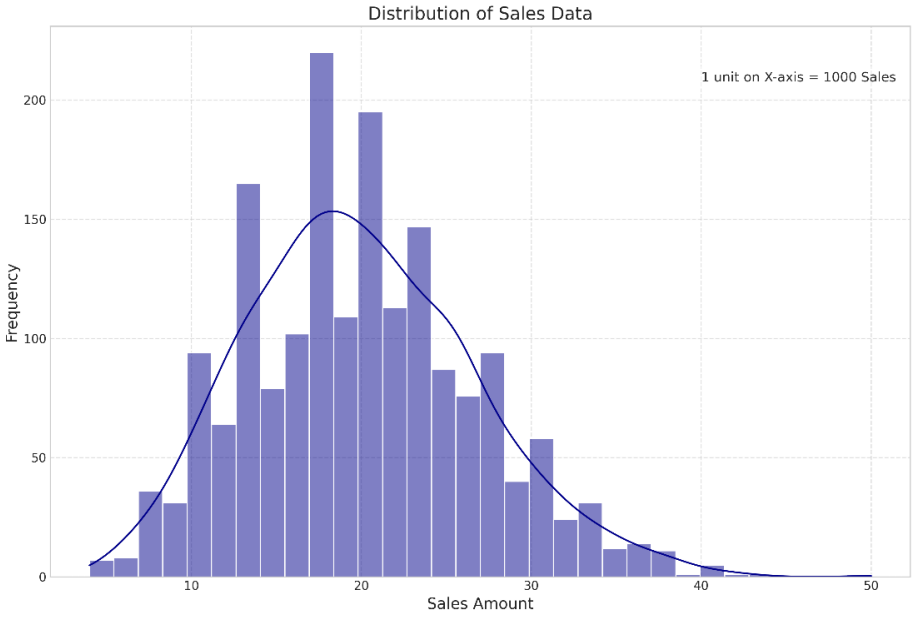
To better understand the range, deviation, and central tendency of the data, we need to develop general statistics for the numerical columns. Among these statistics are the quartiles, standard deviation, mean, and median.

Table 4.1: Data table of "Retail Data Set"

Descriptive Statistics	Value
Total Data	1826
Mean	19.91
Max	32
Min	10
25%	15
50%	19
75%	24
Standard Deviation	6.17

4.4.2.4 Analyzing Target Variables Distribution

Visualizing the pattern of distribution of the objective variable ('sales' in this instance) through the application of histograms or kernel density graphs helps clarify its features and determine skewness or multimodality. This graph combines a histogram and a Kernel Density Estimate (KDE) graph. The frequency distribution of the 'sales' data has been shown through the histogram. The y-axis shows the frequency with which each sales value occurs in the data set, the x-axis represents the range of sales quantities. The KDE plot, a smooth curve, displays the probability density function of the sales data.



. Fig.4.5. Frequency Distribution Sales Graph with Kernel Density Estimate (KDE) plot.

It provides a comprehensive understanding of the distribution's overall shape. This graph shows the distribution's symmetry and the most common sales values (peaks in the histogram and KDE). This graph assists in our comprehension of both the usual sales volume and sales fluctuation.

4.4.2.5 Outlier Detection

Visualizations tools such as box plots have been used to detect expected outliers in the target variable.

4.4.3 Data Cleaning

Data cleaning is an essential process in getting data prepared for evaluation and modelling. This stage concentrated on fixing issues found during exploration to enhance the data quality. This phase includes finding and dealing with missing values, fixing outliers, modifying data types to suitable formats (for example, changing date strings to datetime objects), and to make sure that data is in the correct structure/format for the subsequent processes. This procedure improves data quality, reduces biases, and improves analysis and machine learning model accuracy.

4.4.3.1 Outlier Handling

A box plot demonstrates the distribution of a dataset by showing five important statistical measures: the lowest value, the first quartile (Q1), the median (Q2), the third quartile (Q3), and the highest value. It also shows any outliers. The box moves from Q1 to Q3, which shows the interquartile range (IQR). The median is in the middle of the box. Whiskers proceed from the lowest to the highest values within 1.5 times the IQR. Outliers are data points that are outside of these whiskers.

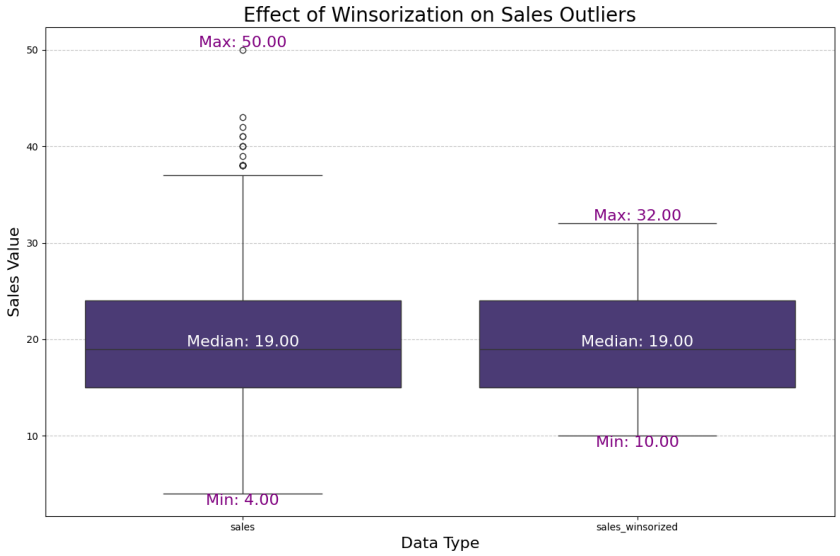


Fig. 4.6. Effect of Winsorization on Removing Sales Outliers

The box plot shows that 50% of the sales values are between about 12 and 30 units, with a median value of about 20. The whiskers show the normal sales range, and there are a few outliers above 38 units, one of which is remarkably high at 50.

4.4.3.2 Feature Engineering

We developed new features to better forecast sales by capturing patterns of time using the 'date' column. We extracted the day of the week, month, year, and day of the year as part of this process. By utilizing these features, it is possible to obtain more information regarding the timing context of each point of data.

4.4.3.3 Dropping Unnecessary Columns

We eliminated several columns that had not been useful for the forecasting mission, such as "date" after feature engineering and "Unnamed: 0," which looked like an index column. These modifications made the dataset less complicated and less noisy.

4.4.3.4 Data Type Verification

We have conducted a final check to ensure that all necessary columns contain numerical data formats suitable for machine learning models. We have implemented type modifications where necessary.

4.4.4 Data Preparation for Modeling

The final procedure entailed transforming the cleaned data into a standard format for training ML models.

4.4.4.1 Feature Scaling

We applied "StandardScaler" to scale all the numerical features except the target variable. This method modifies the features so that their mean is zero and their

variance is one. Scaling is important for many ML methods, especially applied in distance or gradient descent, because it prevents features with higher magnitudes from dominating those with lower magnitudes. This technique makes the models converge faster and perform more efficiently.

4.4.4.2 Data Splitting

We separate the dataset into three separate groups: training, validation, and testing. The training set became 80% of the data, the validation set became 10%, and the testing set became 10%. We applied stratified separating to ensure that the distribution of the target feature ("sales") is equivalent in all three sets. This is very necessary for keeping the data in every division representative.

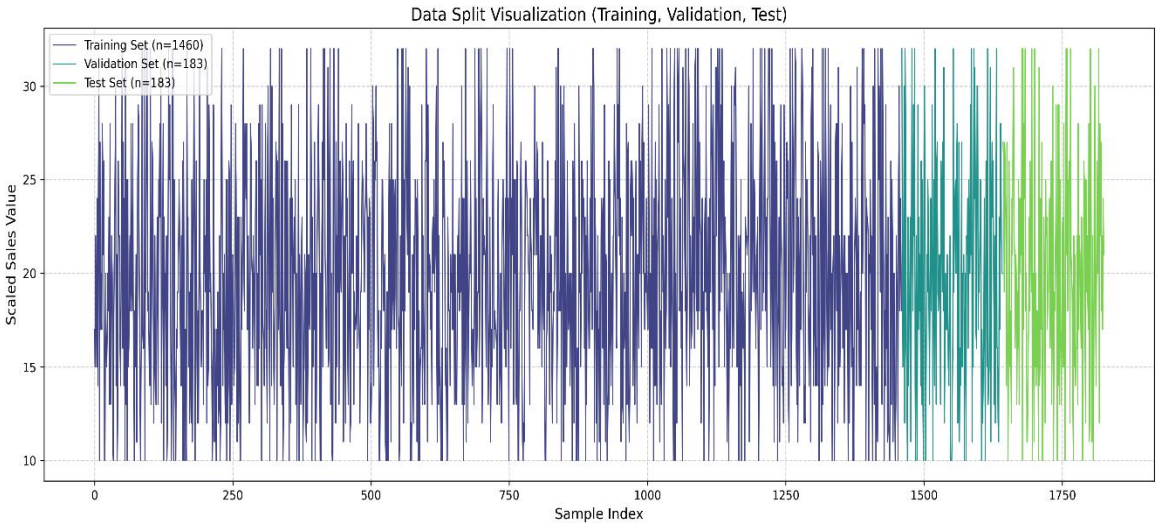


Fig. 4.7. Visualization of Data Split in 80:10:10 Ratio

Training Set: This collection is applied to train or learn ML models.

Validation Set: This is applied to adjust hyperparameters and assess how effectively the model performs throughout development, which protects it from overfitting to the training data.

The testing set: This set has been kept separate from the rest of the model development process and has been employed to present an unbiased assessment of final model performance on new data.

This systematic preparation procedure cleans, structures, and scales data to train and assess ML models for sales forecasting. The graphical explorations and statistical evaluations done during the entire procedure also help us understand the data more effectively and figure out what kinds of issues or structures the models must identify.

4.5 Applied Machine Learning Models

4.5.1 Artificial Neural Network (ANN) & Multi-Layer Perception (MLP)

The fundamental concept behind both ANNs and MLPs is that they learn complex non-linear correlation between input parameters and target outputs by transferring data by using multiple layers of linked "neurons."

Multilayer Perception (MLP) is a feed-forward ANN with at least three layers, trained using backpropagation. The Multi-Layer Perceptron (MLP) is made up of dense layers that are completely interconnected to convert input data from one dimension into another. It is referred to as multi-layer because it consists of an input layer, multiple hidden layers, and a layer for the output.(Zohdi et al.2022).

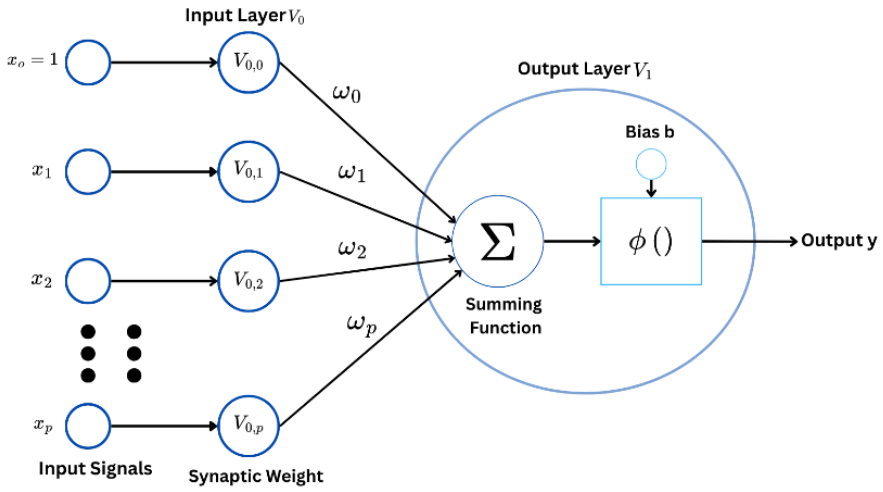


Fig. 4.8. General Architectural View of Artificial Neural Network (López et al., 2022)

1. Input layer: Input layer receives characteristics from the dataset, including various numerical columns and engineered date features (`day_of_week`, `month`, `year`, `day_of_year`). A single input feature is associated with each neuron.

2. Hidden Layer(s): This component processes input information through one or more layers, where neurons develop intricate non-linear correlations with the goal variable (sales) by applying weighted sums and activation functions.

mlp_model = MLPRegressor(...): For the MLP model, the hidden layer is not defined. The MLPRegressor` default uses a hidden layer size with 100 neurons.

ann_model = MLPRegressor (hidden_layer_sizes= (100, 50), ...): In the ANN model, (100, 50) indicates two hidden layers with 100 and 50 neurons, respectively.

3.Output layer: For regression problems, a single neuron having a linear activation function generates forecasts for the target parameter. (sales).

Forward propagation: The process of moving input data from the input layer to the output layer via a network is known as forward propagation.

Weights and Biases: The network learns these parameters during training, and they are related to individual neurons and neuron connections.

$$z = \sum_i w_i x_i + b \tag{4.1}$$

$$Here,\ x_i = Input\ features$$

$$w_i = Corresponding\ Weight$$

b= *Bias*

Activation functions: Activation functions are appiled to add non-linearity to the outputs of neurons in hidden layers or the output layer. ReLU, tanh, and sigmoid are common functions; “relu” and “tanh” are used in the process.

$$Relu\ Funtion(Rectified\ Linear\ Unit): f(z) = \max(0, z)$$

$$Hyperbolic\ tanget: tanh(z) = \frac{2}{1-e^{-2z}} - 1 \quad (4.2)$$

Loss Function: Determines the variance between predicted and actual data.; for regression jobs, Mean Squared Error (MSE) is a popular option. The goal is to reduce this loss.

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2 \quad (4.3)$$

Backpropagation: Backpropagation is a training approach for MLPs that propagates the error backward from the output through the input layer by measuring the gradient of the loss function with respect to each weight and bias.

$$Gradient\ Descent: w = w - \eta.\frac{\partial L}{\partial w} \quad (4.4)$$

$$Here, w\ is\ the\ weight$$

$$\eta = Learning\ rate$$

$$\frac{\partial L}{\partial w} = gradient\ of\ the\ loss\ function\ with\ respect\ to\ the\ weight$$

Optimize: The optimizer is an algorithm (Adam or Stochastic Gradient Descent, or SGD) that uses gradients from backpropagation to update weights and biases to decrease the loss function. Both "adam" and "sgd" are used in the program.

$$SGD: w = w - \eta.\frac{\partial L}{\partial w} \quad (4.5)$$

$$Momentum: m_t = \beta_1 m_{t-1} - (1-\beta_1).g_t \quad (4.6)$$

$$Adaptive\ learning\ rate: v_t = \beta_2 v_{t-1} - (1-\beta_2).{g_t}^2 \quad (4.7)$$

Here, g_t represents the gradient at time t, β_1, β_2 are decay rates.

The training process consists of iterative training across several epochs, during which the network computes loss, does forward propagation, and optimizes and

backpropagates parameters. To avoid overfitting, the "early_stopping=True" parameter in "MLPRegressor" stops training if validation performance does not get better after a predetermined number of epochs. (GeeksforGeeks, 2025)

4.5.2 Gradient Boosting

Gradient boosting is an ensemble-supervised machine learning algorithm. that applies to tasks such as regression and classification. It is defined by the continuous training of several models with an emphasis on resolving errors from prior forecasts. This method pays special attention to tough cases by giving more importance to instances where predictions were wrong, which helps to gradually lower the error, much like how artificial neural networks work. In gradient boosting, after creating weak models, the predictions are checked against the real outcomes to see how wrong they were, and this information is then used to figure out how to adjust the model's settings to avoid making the same mistakes again. Gradient boosting uses hyperparameters such as random forests and learning rate and loss functions to improve predictions over single-model methods. (Masui, 2024)

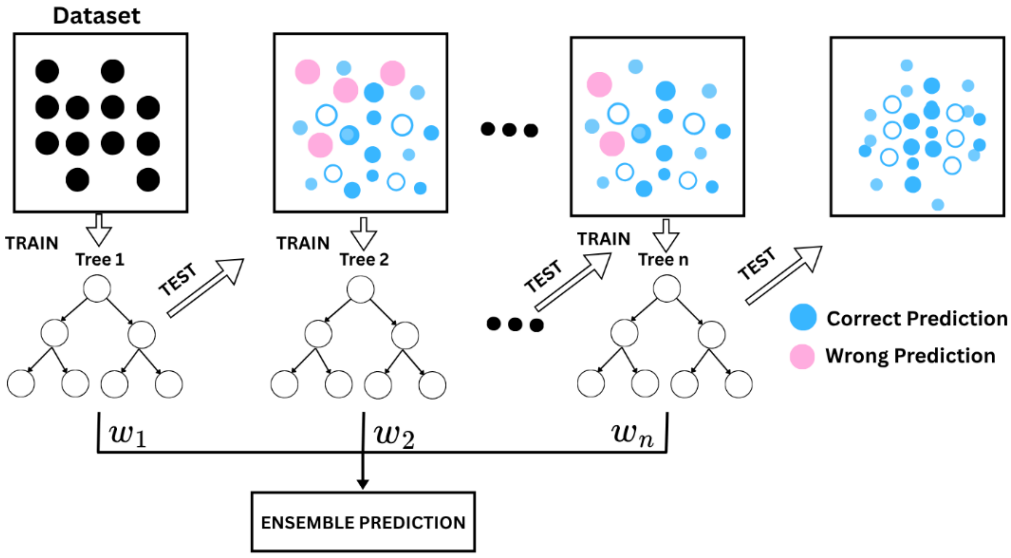


Fig. 4.9. Gradient Boosting Algorithm Workflow (Zhang et al., 2021)

ALGORITHM 4.1: GRADIENT BOOSTING ALGORITHM

1. Initializing Model with Constant Value

$$F_0(x) = arg_\gamma min \sum_{i=1}^{n} L(y_i, \gamma)$$

2. For m=1 to M:

Compute Residuals $\gamma_{im} = -\left[\frac{\partial L\left(y_i, F(x_i)\right)}{\partial F(x_i)}\right]_{F(x)=F_{m-1(x)}} \quad for\ i = 1 \ldots . n$

Train regression tree with feature x against γ and create terminal node reasons $R_{jm} \quad for\ j = 1 \ldots j_m$

Compute $\gamma_{jm} = arg_\gamma min \sum_{x_i \epsilon R_{jm}} L(y_i F_{m-1}(x_i) + \gamma) \quad for\ j = 1 \ldots j_m$

Update the model,

$$F_m(x) = F_{m-1}(x) + v \sum_{i=1}^{jm} \gamma_{jm} 1(1 \epsilon R_{jm})$$

4.5.3 Random Forest

A random forest is a strong machine learning algorithm applied for classification and prediction, dependent on immense amounts of high-quality data for training. This method enhances the system's effectiveness by optimizing algorithms, improving software and hardware productivity, and allowing superior decision-making and forecasting. This technique generates several decision trees from random groups of data, combines their results into final forecasts, and addresses unbalanced and missing data while minimizing overfitting. The approach is recognized for its efficiency and adaptability; it is suitable for automatic categorization, data prediction, and supervised learning while encouraging creativity through multidisciplinary research and varied data processing approaches. (Salman et al., 2024)

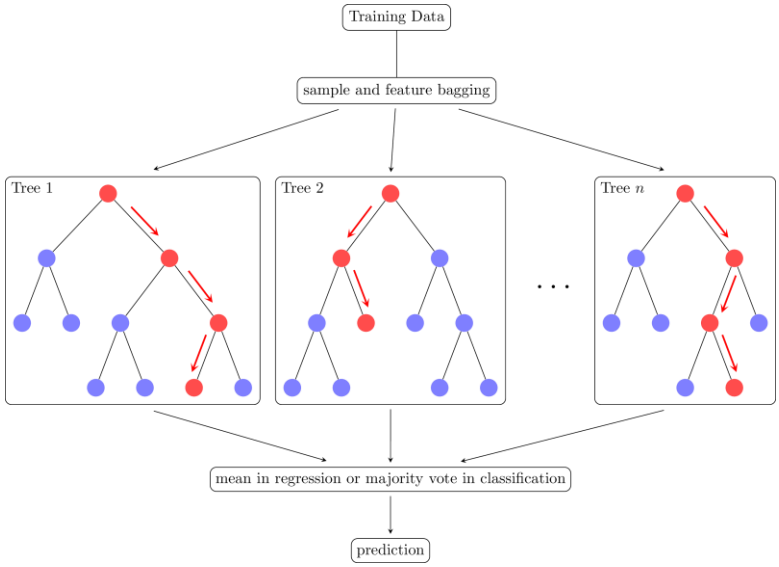


Fig. 4.10. Random Forest Algorithm Workflow (Riebesell, 2022a)

4.5.4 Support Vector Regressor (SVR)

Support Vector Machine (SVM) has been customized for regression problems, known as Support Vector Regression (SVR), while keeping its key features, especially the principle of maximal margin. SVR functions on concepts related to SVM for classification; however, it encounters specific difficulties caused by the constant flow of real-number outcomes, complicating forecasts due to their endless possibilities. SVR addresses those challenges by introducing an epsilon margin of tolerance, which enables a certain degree of error. Despite the increased complexity of the algorithm, its main goal remains to reduce error by detecting a hyperplane that optimizes the margin and allows for a predetermined degree of error tolerance. (*Support Vector Regression*, n.d.)

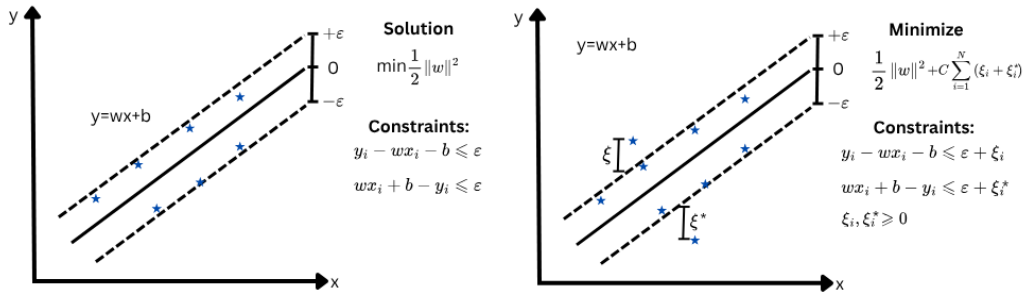


Fig. 4.11. Linear Support Vector Regression Working Process

Linear SVR equation:

$$y = \sum_{i=1}^{N}(a_i - a_i^*).\langle x_i, x\rangle + b \quad (4.8)$$

Non-linear SVR equation: The implementation of kernel functions carries out the linear split by transforming the data into a feature space with higher dimensions.

$$y = \sum_{i=1}^{N}(a_i - a_i^*).\langle \varphi(x)_i, \varphi(x)\rangle + b$$

$$y = \sum_{i=1}^{N}(a_i - a_i^*).K(x_i, x) + b \quad (4.9)$$

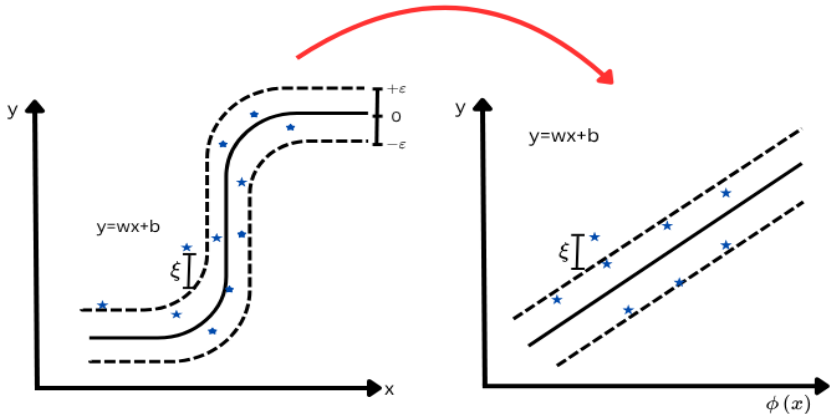


Fig. 4.12. Non-linear Support Vector Regression Working Process

Karnel Function:

Polynomial: K (x_i, x_j)=$(x_i, x_j)^d$ (4.10)

Gaussian Radial Basis Function: K (x_i, x_j)=exp $(-\frac{\|x_i - x_j\|^2}{2\sigma^2})$ (4.11)

4.5 Proposed Architecture

In the following flow diagram illustrates a comprehensive and systematic procedure for constructing an efficient sales forecast system that employs both single and hybrid machine learning models. The initial stage in the process is data preprocessing, which means collecting the dataset and extracting relevant features (date-based elements like day, month, and year). After extracting the features, we apply StandardScaler to scale the numerical data so that all features are on the same scale. This stage is a vital phase for models like SVR and neural NN. After successfully completing the preprocessing phase, we split the dataset into three distinct parts with an 80:10:10 ratio for training, validation, and testing purposes.

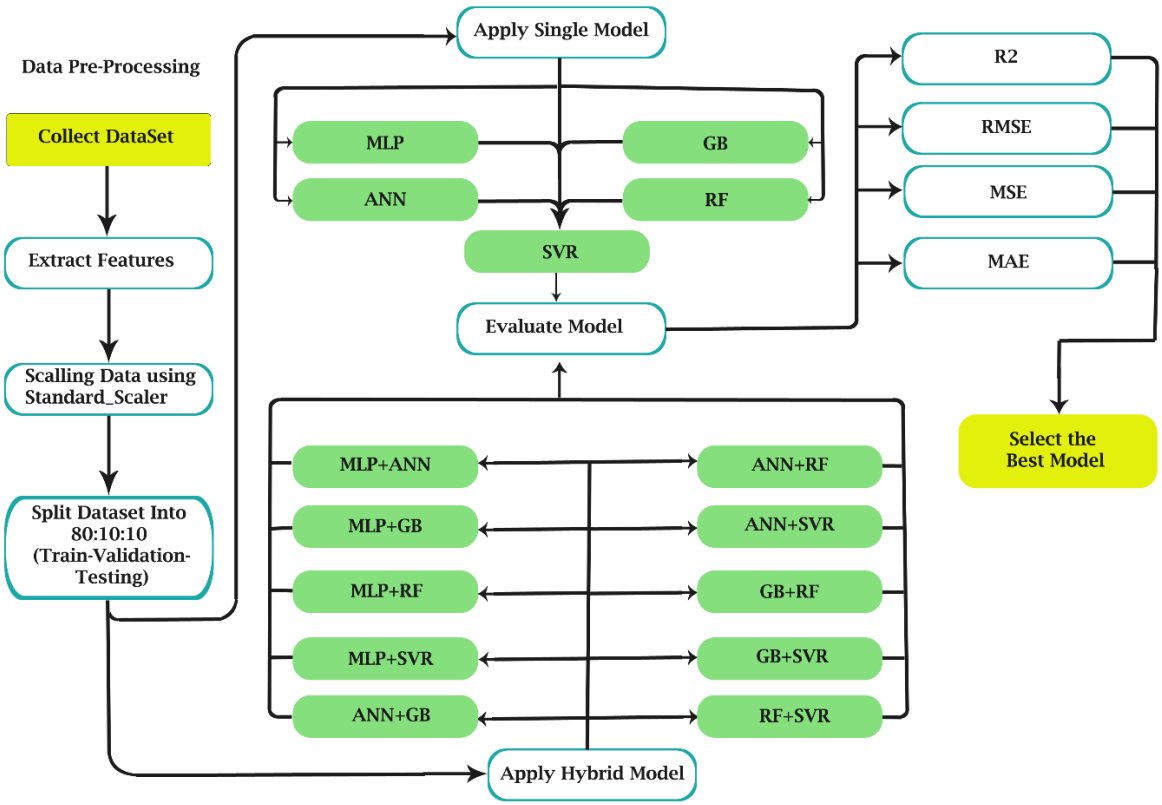


Fig. 4.12. The Workflow of the Proposed Architecture

By ensuring that every subset keeps the target variable's distribution, this classified split provides a balanced ground for model training and assessment. Certain preparation the

data, certain ML models, such as MLP (Multilayer Perceptron), ANN, SVR, GB and RF have been applied. After training all the models, we evaluated their performance by applied various statistical performance metrics such as R2 (coefficient of determination), RMSE (Root Mean Squared Error), MSE (Mean Squared Error), and MAE (Mean Absolute Error). These metrics provide in-depth insights into model accuracy and error size, and they also help evaluate how well each model fits this dataset. To further enhance the accuracy of predictions, we developed various combinations of pairwise hybrid models. In this approach, two models are combined (such as MLP+ANN, ANN+RF, or GB+SVR) to leverage their complementary strengths. For instance, we use the non-linear learning power of neural networks with the ensemble strength of tree-based models. Then we evaluate those hybrid models by using similar evaluation matrices, enabling direct comparisons with individual models. Lastly, we have chosen the top-performing model, considering the results of both single and hybrid model analyses. We assume that based on the unseen data, our model will generate the most accurate and applicable forecasts. This systematic and continuous workflow enhances accurate forecasting and facilitates trustworthy decision-making in business environments.

4.7 Summary of the Chapter

The datasets that were used in this research, data preprocessing techniques, feature extraction techniques, and proposed architecture have all been explained in this chapter.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 Overview

This section explains how to evaluate the suggested various types of machine learning models, both single & hybrid models. Those models will forecast sales using retail datasets and show their accuracy, validity, and supply chain forecasting abilities.

5.2 Experimental Setup

This experiment has been performed on a laptop that was equipped with a 64-bit Windows operating system, 8 GB of RAM, and an Intel Core i5-8265U CPU that operated at a speed of 1.60 GHz to 1.80 GHz. However, the proposed machine learning models required more computational power than the local machine could efficiently handle. Therefore, Google Collaboratory, a free, cloud-based platform provided by Google, was used to run experiments. It offers a high-performance environment suitable for training machine learning models and functions similarly to Jupyter Notebook.

5.3 Evaluation Metrics

In this study, four distinct types of evaluation matrices have been used. Here we explain all the matrices.

5.3.1 Root Means Square Error (RMSE)

Root Mean Square Error is a statistical measure used to assess the correctness of models, such as regression models. It calculates the mean variance between predicted and actual data and reports its effect on the data. (Chavali et al., 2023)

$$RMSE = \sqrt{\frac{\sum_i^N (x_i - \hat{x_i})}{N}} \quad (5.1)$$

Here, i= Variable i x_i = *Actual time series*

$\hat{x_i}$ = *Predicted time series*

N=Number of non-missing data

5.3.2 Mean Absolute Error (MAE)

Mean Absolute Error is an important way to measure how well a regression model works. A lower MAE means a more effective model. It finds a distinction between two continuous variables, X and Y, by applying the given equation.. (Chavali et al., 2023)

The formula is:

$$MAE = \frac{\sum_{i=1}^N |y_i - x_i|)}{n} \quad (5.2)$$

Here, i= Variable i y_i = Predicted Data

x_i= Actual Data

N=Number of non-missing data

5.3.3 Coefficient of Determination or R^2 Score

The R^2 score indicates how much of a dependent variable's variance a model explains. It is important for figuring out how dispersed the data is and showing slight variations between the actual and predicted data. When the R^2 number is high, there doesn't seem to be much difference between the actual and forecast values.. Eighty percent of the variability of the dependent variable can be explained by the independent variable, according to an R^2 value of 0.8. (Tanvir et al.,2024) .The formula is:

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)}{\sum(y_i - \bar{y}_i)} \tag{5.3}$$

Here, $y_i = The\ y\ value\ for\ observation\ i$

$\hat{y}_i = Predicted\ value\ of\ y\ for\ observation$

$\bar{y}_i = The\ mean\ of\ y\ value$

5.3.4 Mean Squared Error (MSE)

The mean squared error (MSE) is the difference between actual and predicted mean value in a statistical analysis. It is important to square differences since certain data values may be higher than predicted, resulting in positive differences and others negative. Squaring these differences prohibits zero differences because measurements are equally likely to exceed forecast values. (Stewart et al.,2023). The formula is:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{5.4}$$

$y_i = Actual\ Values$

$\hat{y}_i = Predicted\ Values$

$n = Number\ of\ Data\ Point$

5.4 Performance Analysis on "Single Model "

Table 5.1 demonstrates how well five different ML models—ANN, MLP, GB, RF, and SVR—did on the "Retail Data Set," applying different standard regression metrics such as MSE, RMSE, MAE, and R^2. This research analysis demonstrates that Random Forest (RF) provides significantly better results than all the other models in most criteria.

Table 5.1: Performance Measure of "Retail Data Set" Using all Evaluation Matrices for all Single Models.

Classifier	MSE	RMSE	MAE	R^2
ANN	28.0499	5.2962	4.3204	0.2450
MLP	28.0499	5.2962	4.3204	0.2450
GB	19.7229	4.4411	3.5082	0.4691
RF	**16.8087**	**4.0998**	**3.3014**	**0.5476**
SVR	17.5447	4.1886	3.3816	0.5277

Specifically, RF has the lowest MSE (16.8087), RMSE (4.0998), and MAE (3.3014), as well as the highest R^2 score (0.5476), which means it performs more effectively at making predictions. On the other hand, ANN and MLP provide the same poor outcomes, with the highest error values (MSE = 28.0499, RMSE = 5.2962) and the lowest R^2 (0.2450). The result shows that both models have trouble finding the patterns in the retail dataset. In summary, the analysis highlights the effectiveness of ensemble methods, like Random Forest (RF), in handling complex retail data for accurate sales forecasting in this research.

5.5 Performance Analysis on "Retail Data Set" in Demand Forecasting (Single Model Evaluation)

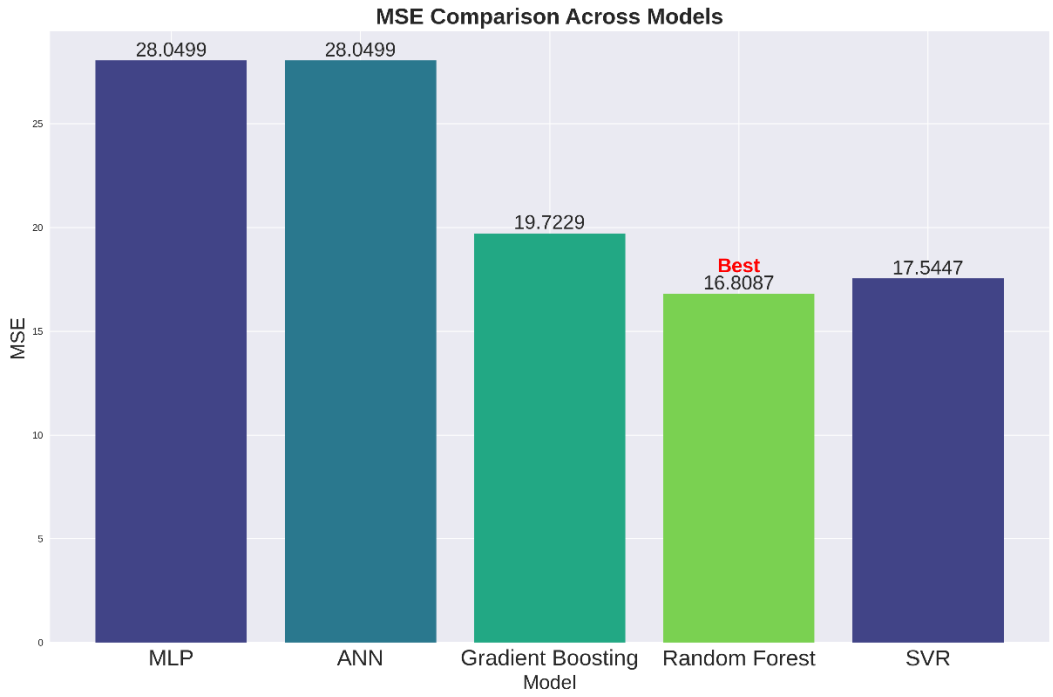


Fig. 5.1. MSE Comparison Across all Single Models

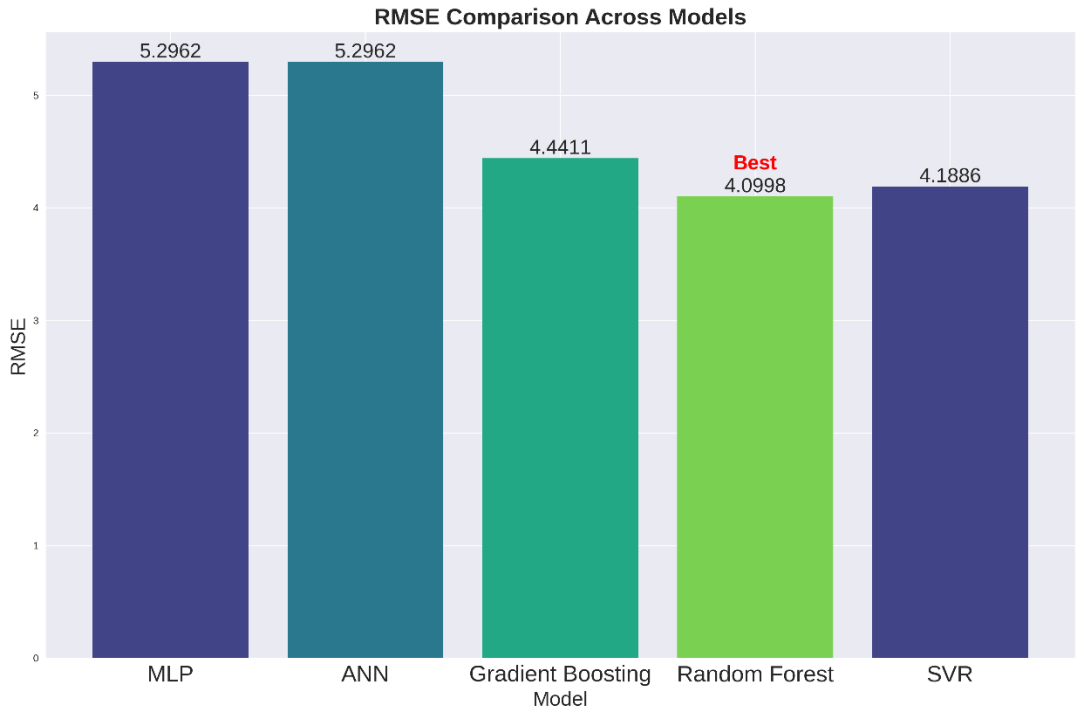


Fig. 5.2. RMSE Comparison Across all Single Models

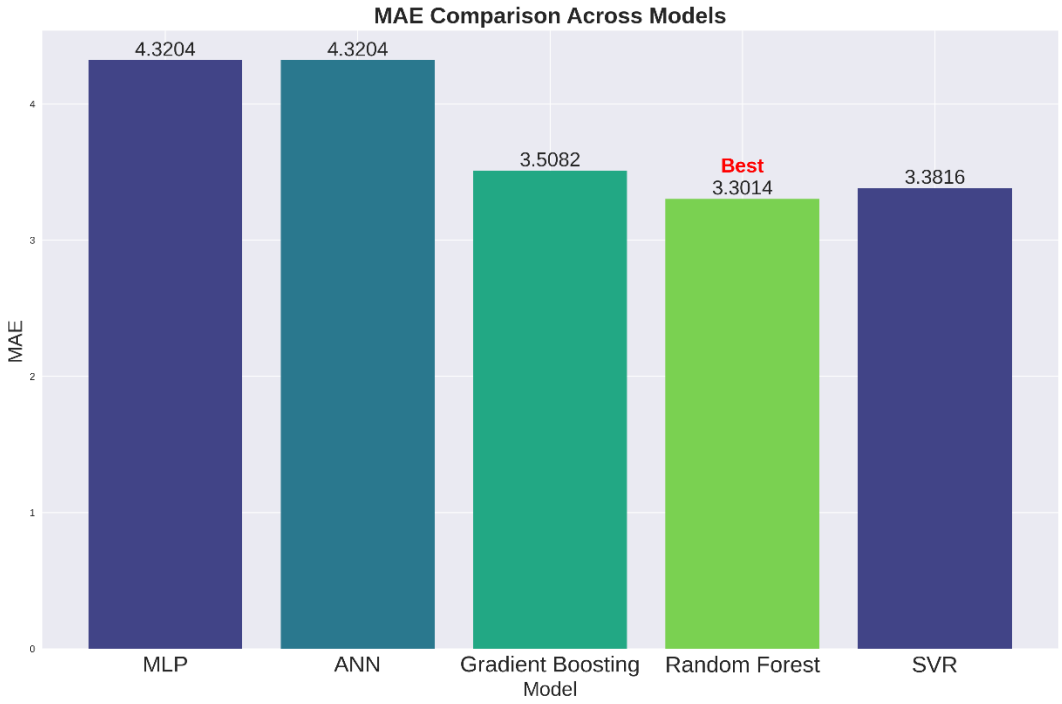


Fig. 5.3. MAE Comparison Across all Single Models

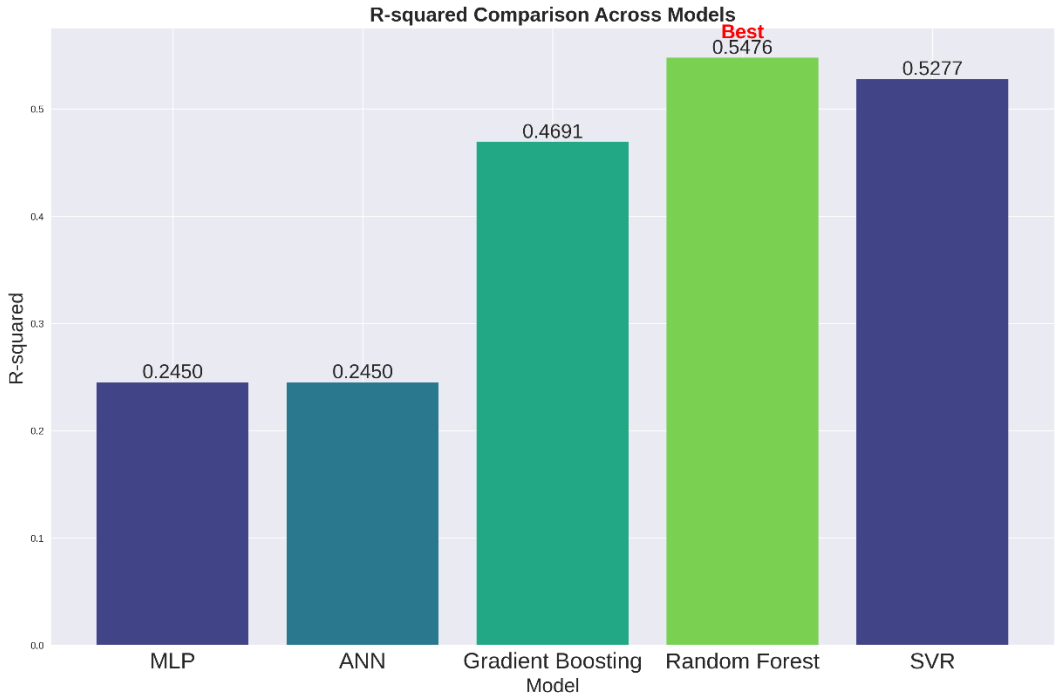


Fig. 5.4. R-squared Comparison Across all Single Models

5.6 Actual vs. Predicted Sales Comparison of Single Models

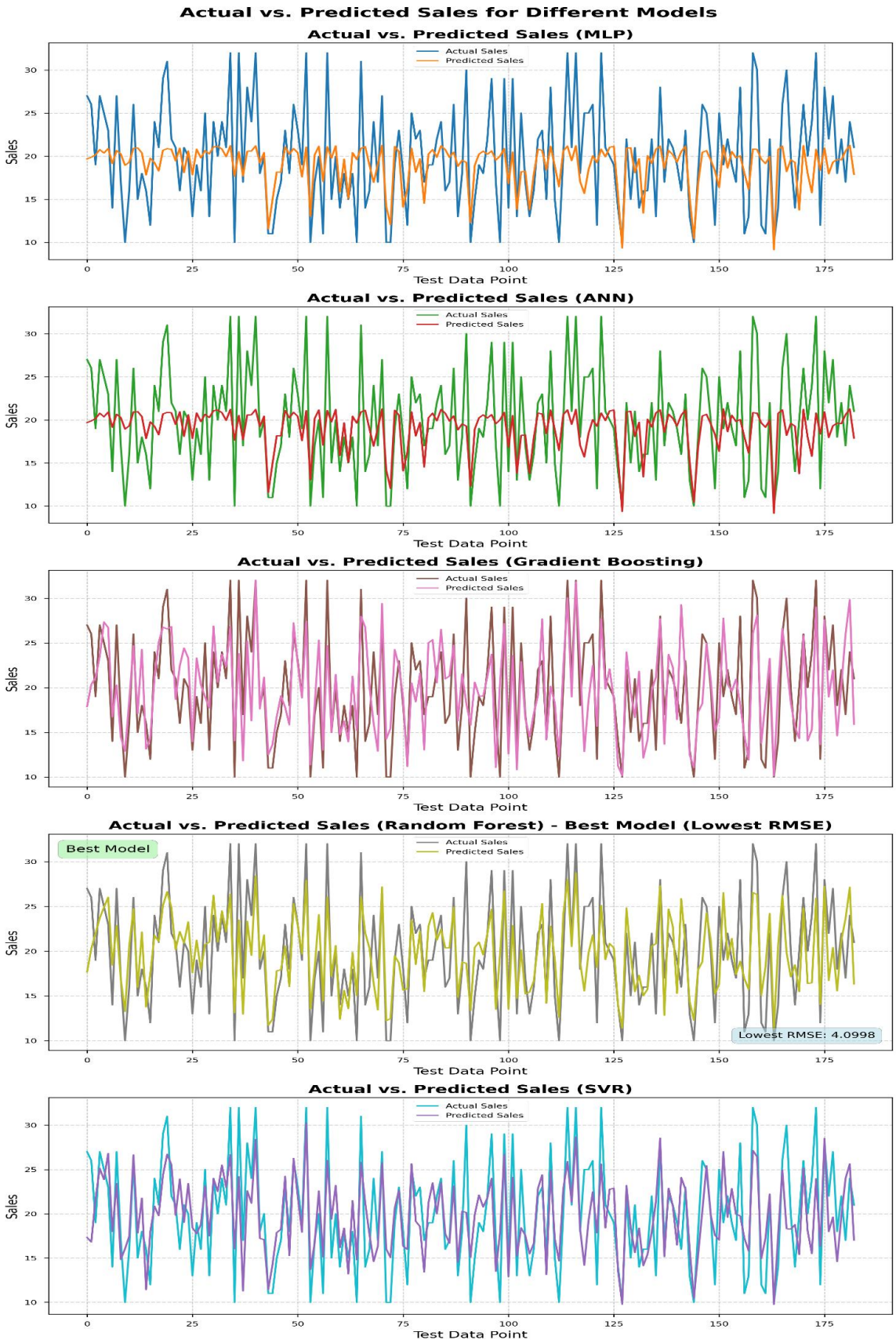


Fig 5.5. Actual vs. Predicted Sales Comparison Across all Single Models on the Test Dataset.

5.7 Performance Analysis on "Hybrid Model"

Table 5.2 shows a detailed comparison of different hybrid machine learning models based on four significant metrics: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and the coefficient of determination (R^2). These metrics are crucial for figuring out how accurate and reliable the models are at making predictions. The hybrid model RF+SVR is the best because it has the lowest error values among those metrics (MSE of 16.4851, RMSE of 4.0602, MAE of 3.2634). It also achieved the greatest R^2 value of 0.5563, which means it performed very effectively in this dataset.

Table 5.2: Performance Measure of "Retail Data Set" Using all Evaluation Matrices for all Hybrid Models.

Classifier	MSE	RMSE	MAE	R^2
MLP+ANN	28.0499	5.2962	4.3204	0.2450
MLP+GB	19.9585	4.4675	3.5802	0.4628
MLP+RF	20.0116	4.4734	3.5911	0.4613
MLP+SVR	20.4097	4.5177	3.6461	0.4506
ANN+GB	19.95885	4.4675	3.5802	0.4628
ANN+RF	20.0116	4.4734	3.5911	0.4613
ANN+SVR	20.4097	4.5177	3.6461	0.4506
GB+RF	17.3469	4.1686	3.3537	0.5323
GB+SVR	17.1448	4.1406	3.3270	0.5385
RF+SVR	**16.4851**	**4.0602**	**3.2634**	**0.5563**

GB+SVR and GB+RF are two other important models, with MSE values of 17.1448 and 17.3469, respectively. Both models have low RMSE and MAE outcomes and R^2 values above 0.52, which means they are good at making predictions in retail settings. GB+SVR achieves a good blend of accuracy and generalization, with an R^2 of 0.5385. On the other hand, the MLP+ANN hybrid model performed the worst, with a maximum MSE of 28.0499 and a minimal R^2 of 0.2450, which means it had not been particularly good at ensuring predictions or being reliable. Similarly, models such as MLP+RF and ANN+RF exhibit intermediate error ranges.

5.8 Performance Analysis on "Retail Data Set" in Demand Forecasting (Hybrid Model Evaluation)

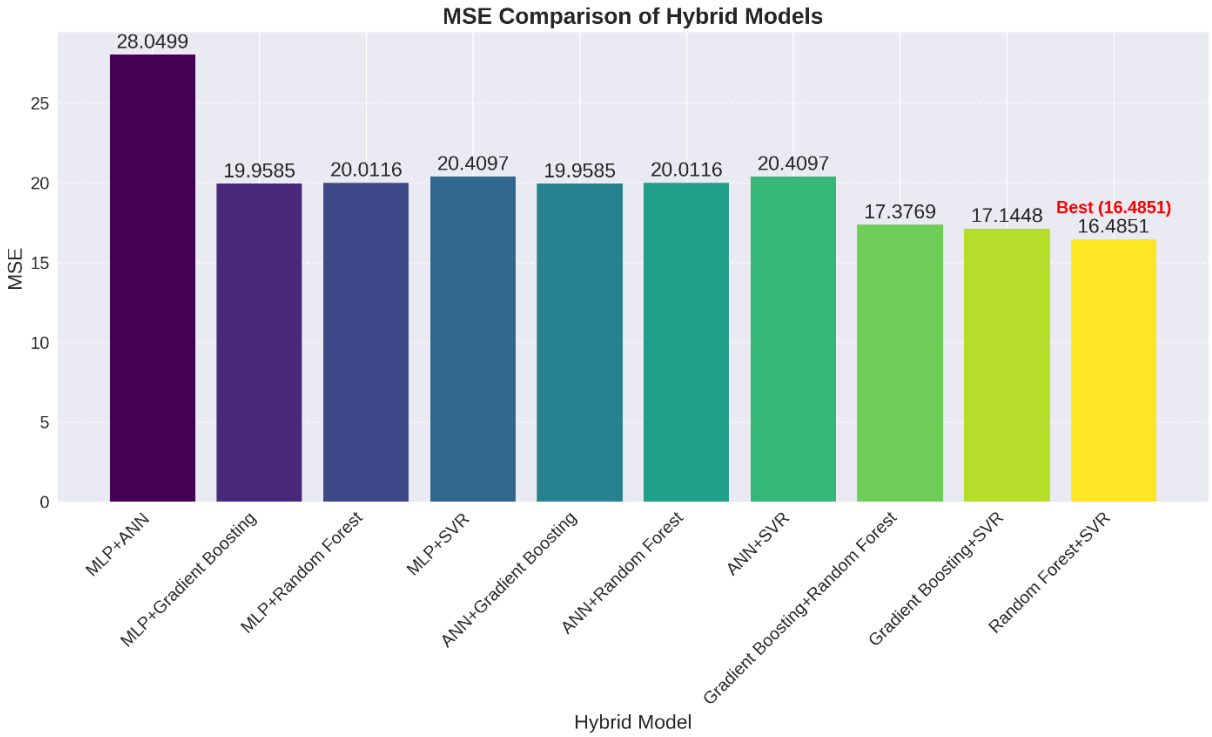


Fig. 5.6. MSE Comparison Across all Hybrid Models

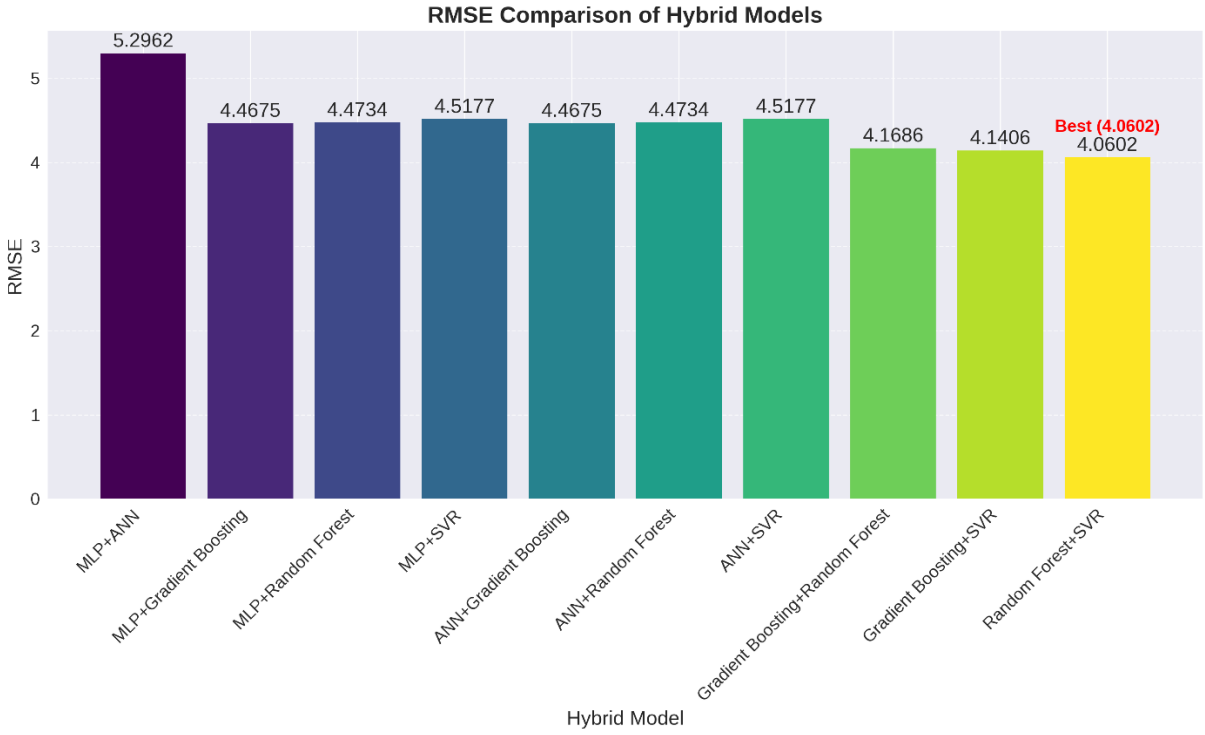


Fig. 5.7. RMSE Comparison Across all Hybrid Models

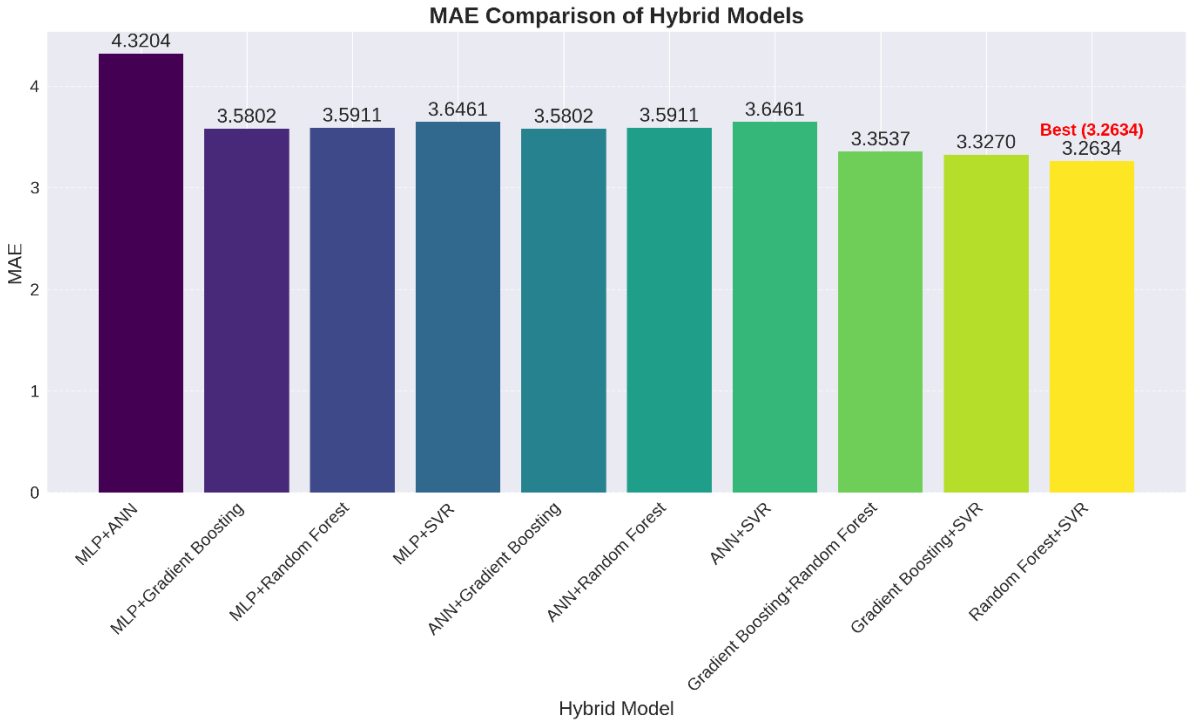


Fig. 5.8. MAE Comparison Across all Hybrid Models

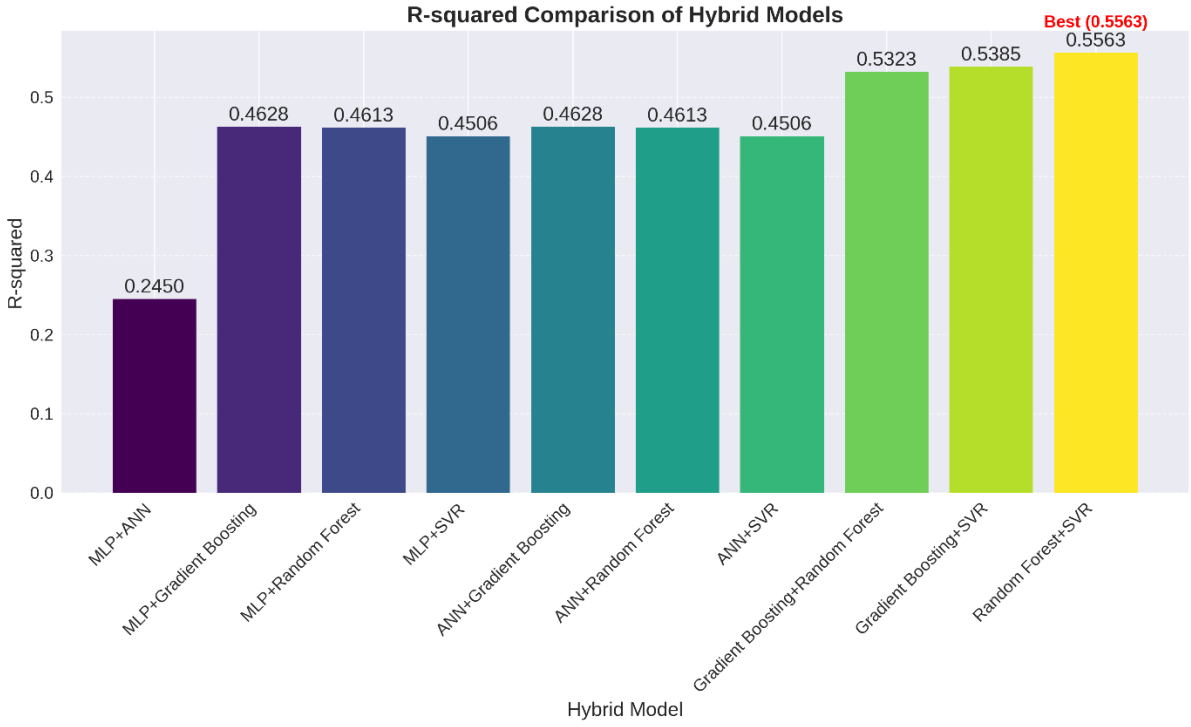


Fig. 5.9. R-squared Comparison Across all Hybrid Models

5.9 Actual vs. Predicted Sales Comparison of Hybrid Models

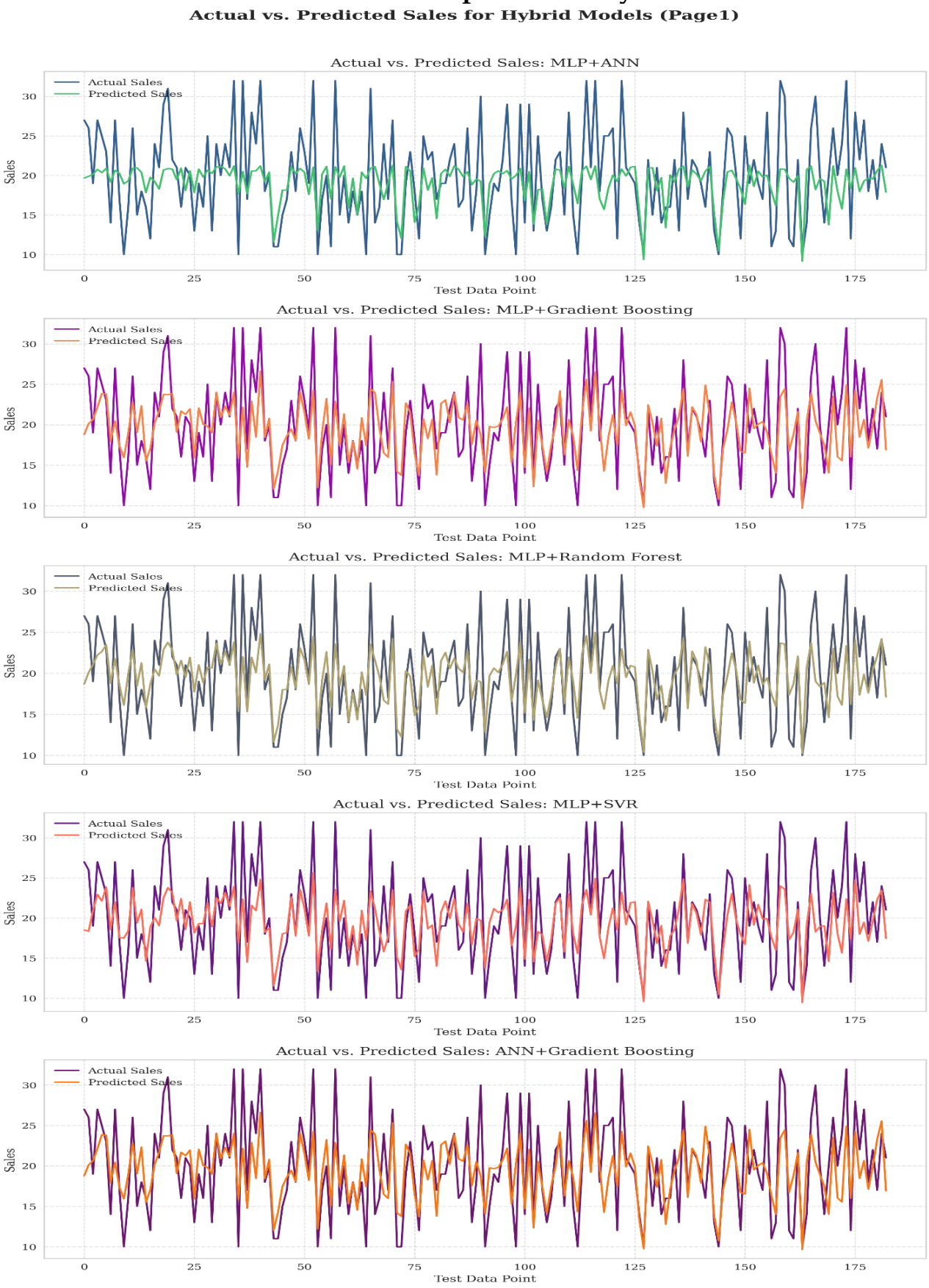


Fig. 5.10. Actual vs. Predicted Sales Comparison Across First Five Hybrid Models on the Test Dataset.

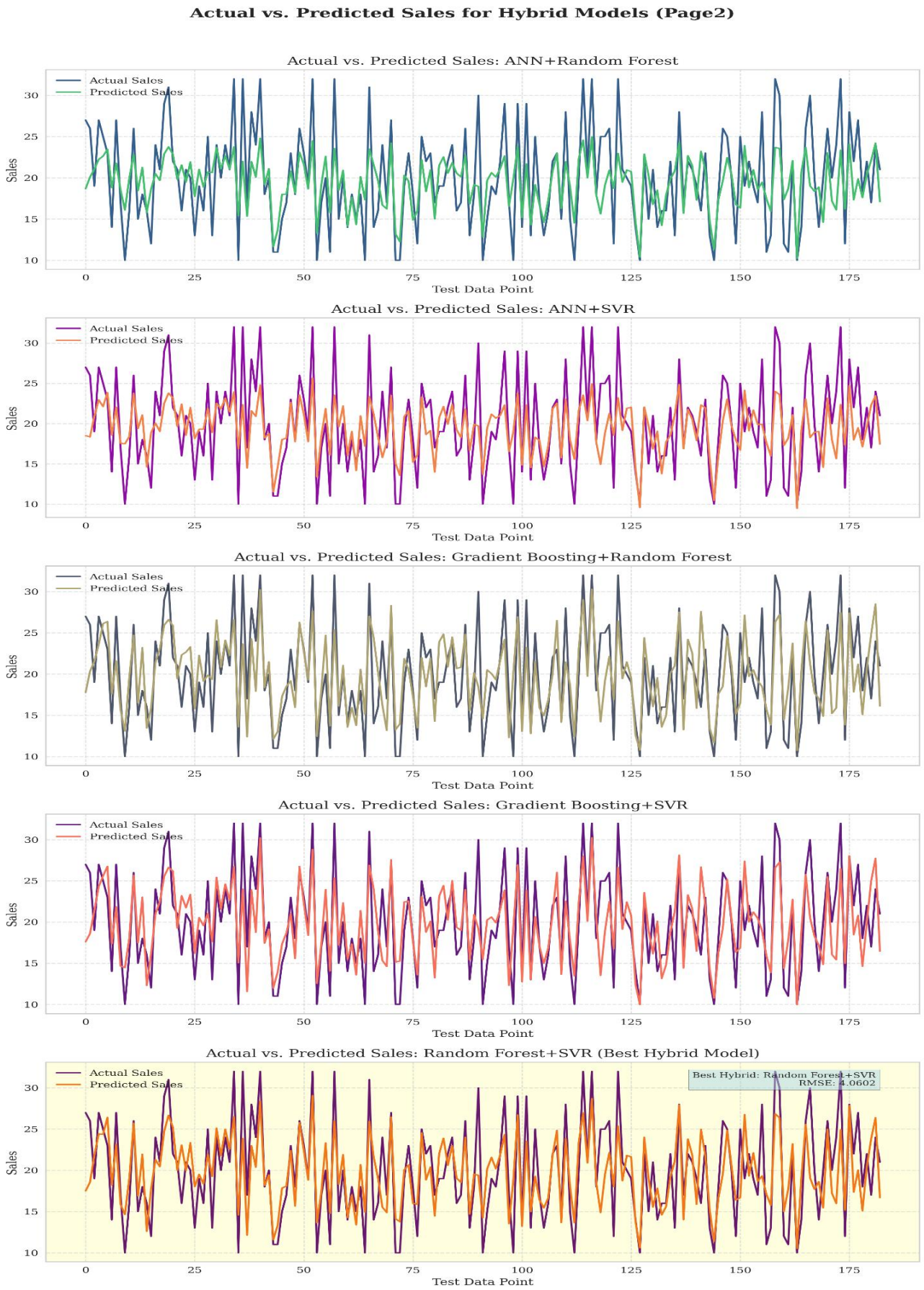


Fig 5.11. Actual vs. Predicted Sales Comparison Across Last Five Hybrid Models on the Test Dataset

Table 5.3: Summary of Best Outcomes Achieved by Single and Hybrid Learning Models

Best Model	MSE	RMSE	MAE	R^2
Single Model (RF)	**16.8087**	**4.0998**	**3.3014**	**0.5476**
Hybrid Model (RF+SVR)	**16.4851**	**4.0602**	**3.2634**	**0.5563**

5.10 Summary of the Chapter

In this chapter, we covered the evaluation criteria that have been employed to confirm the best performance of the supply chain demand forecasting model, both single and hybrid models.

CHAPTER 6

CONCLUSIONS

6.1 Conclusion

This study analyzed the efficient implementation of machine learning to improve demand prediction accuracy in the retail supply chain sector, especially compared to single and hybrid ML models. This study applied data from a Bangladeshi retailing company to evaluate five separate models (ANN, MLP, GB, RF, and SVR) and ten hybrid models, which are combinations of two of these models. The evaluation provides important details about how well the model operates in real-life retail predicting situations by using RMSE, MAE, MSE, and R2 metrics. The Random Forest (RF) model has achieved the highest overall performance of all single models, with an RMSE of 4.0998, an MAE of 3.3014, and an R^2 of 0.5476. With an RMSE of 4.0602, an MAE of 3.2634, and an R^2 of 0.5563, the hybrid RF+SVR model performed better than all of the other models. This research suggests that hybrid models increase accurate forecasting and flexibility by combining the strengths of multiple models. The results indicate that hybrid models are better at determining the seasonal and nonlinear sales patterns than single models. The study also demonstrates that it is possible and important to use localized datasets in machine learning experiments, especially in emerging countries where public data is difficult to collect. In conclusion, this research confirms that hybrid machine learning models could make it much easier to estimate sales demand in the retail supply chain. These results complement the growing body of data on AI-driven forecasting, providing the foundation for businesses and researchers to build more intelligent, context-aware demand planning tools.

6.2 Major Findings and Future Scopes

The key findings of the study are summarized as follows:

a) The Random Forest model achieved the best overall, with an MSE of 16.8087, an RMSE of 4.0998, an MAE of 3.3014, and an R^2 of 0.5476. This result indicates that the model is extremely accurate at demand forecasting.

b) ANN and MLP were the poorest single models, with MSE: 28.0499, RMSE: 5.2962, MAE: 4.3204, and R^2: 0.2450, which means they failed to perform an adequate job of generalizing to the dataset.

c) The RF + SVR hybrid model performed better than all others, as proven by its MSE of 16.4851, RMSE of 4.0602, MAE of 3.2634, and R2 of 0.5563.

d) The GB + SVR and GB + RF hybrid models appeared to be two more efficient options that outperformed the initial models and produced more reliable predictions.

e) Hybridization made predictions more accurate by combining the strengths of different algorithms, specifically for finding nonlinear sales patterns.

f) This study suggests that using real-world data to generate ML-based forecasts, especially with hybrid models, is an appropriate strategy to improve demand forecasting in new retail sectors.

There are many ways that this research can be expanded in the future to make it more beneficial and efficient. Adding new variables, like holidays, promotional offers, or weather effects, may assist the model in making better predictions. It may be possible to implement more advanced deep learning methods, like LSTM or Transformer networks, to deal with sales patterns that are more complex and last longer. A real-

time forecasting system that keeps updating forecasts will be beneficial because it allows businesses to make decisions instantaneously.

Furthermore, developing the model to be simpler to understand may assist individuals in trusting and embracing the forecasts. Lastly, this approach can be employed in other fields where predicting demand is essential, including medical care, manufacturing, or financial services.

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